

Technologies for E-Commerce

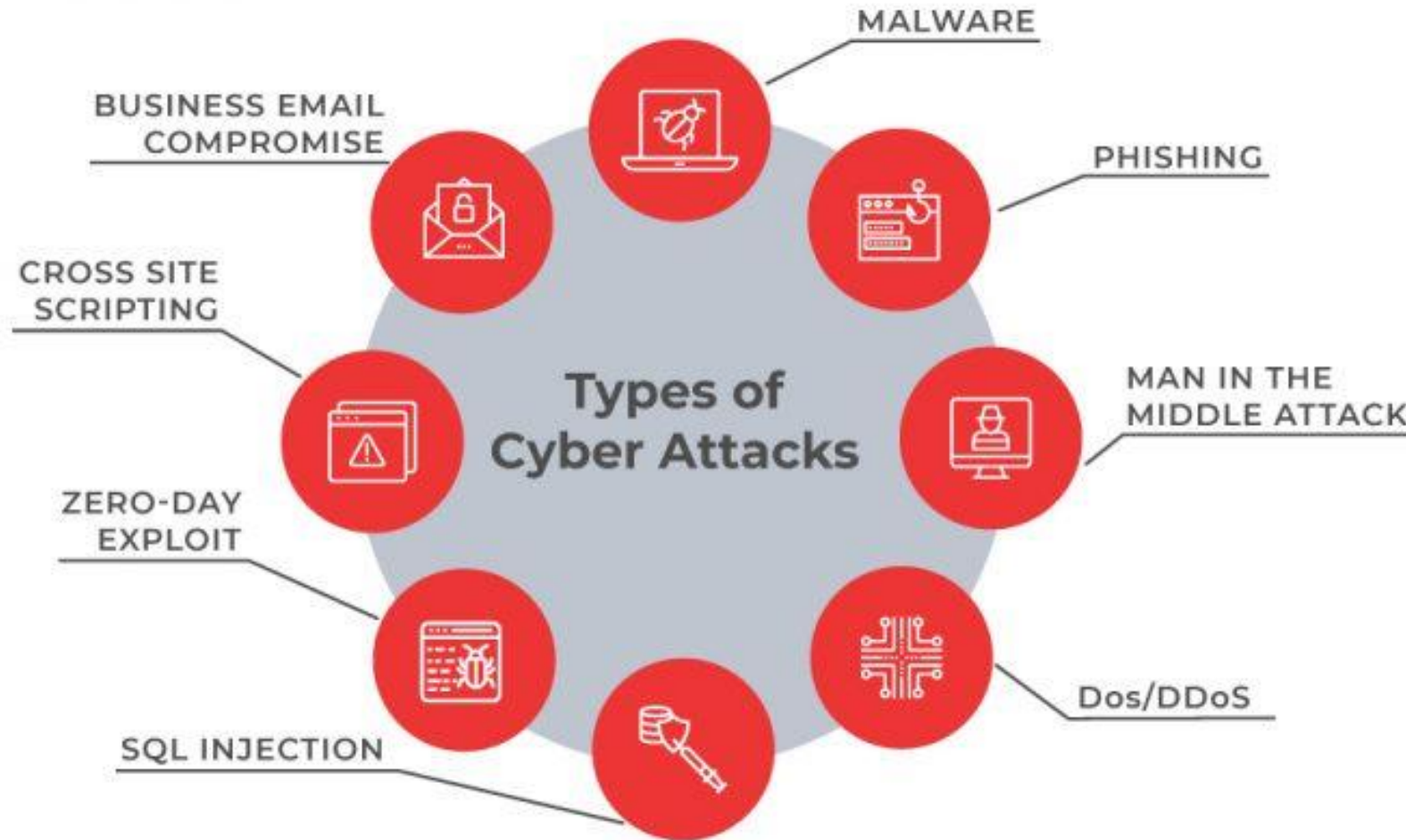
CAN302

**Department of Communications and Networking
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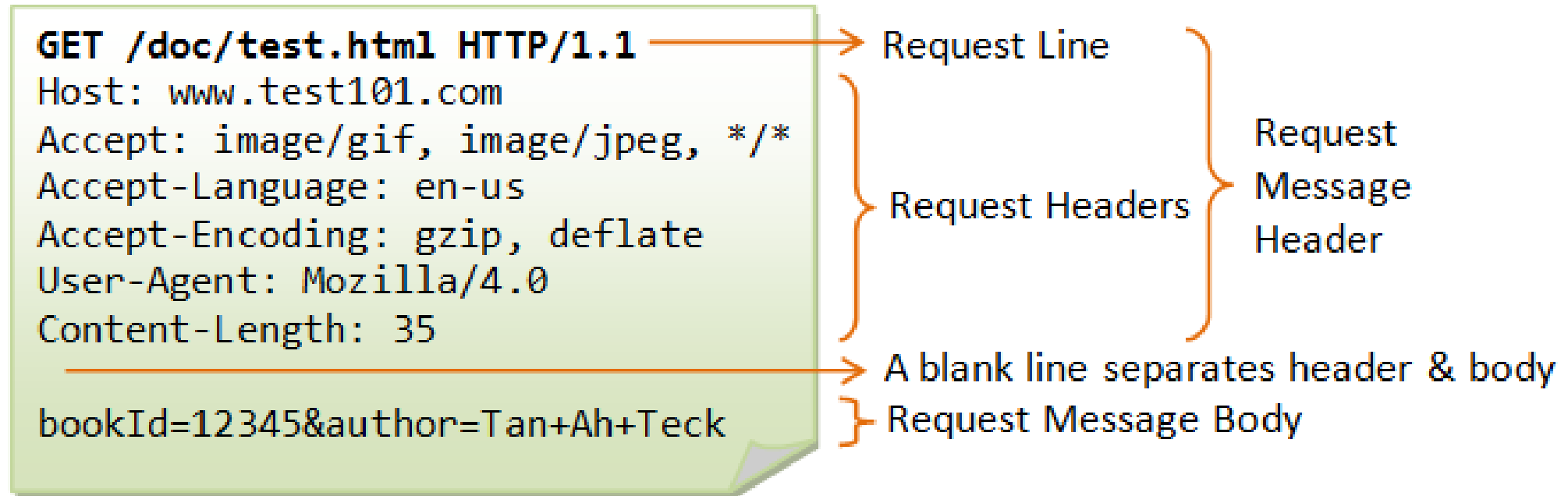
Lecture 7 – Introduction to cryptography



Typical cyber attacks

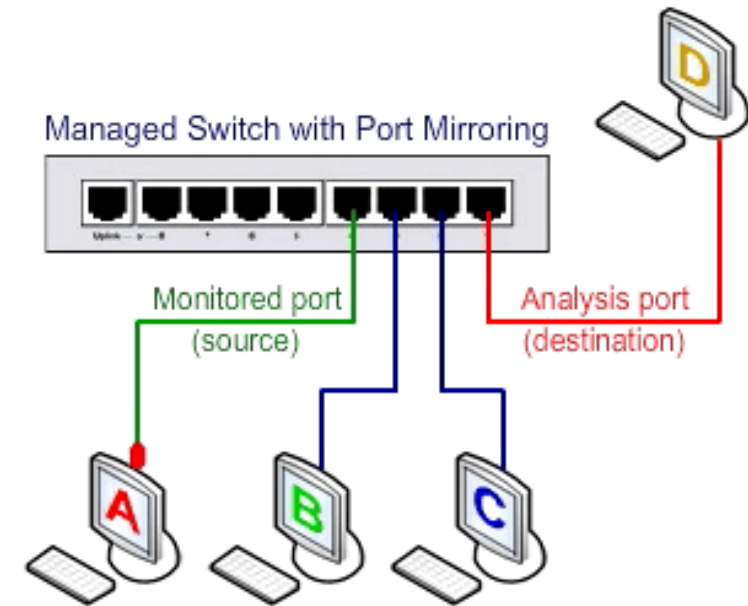
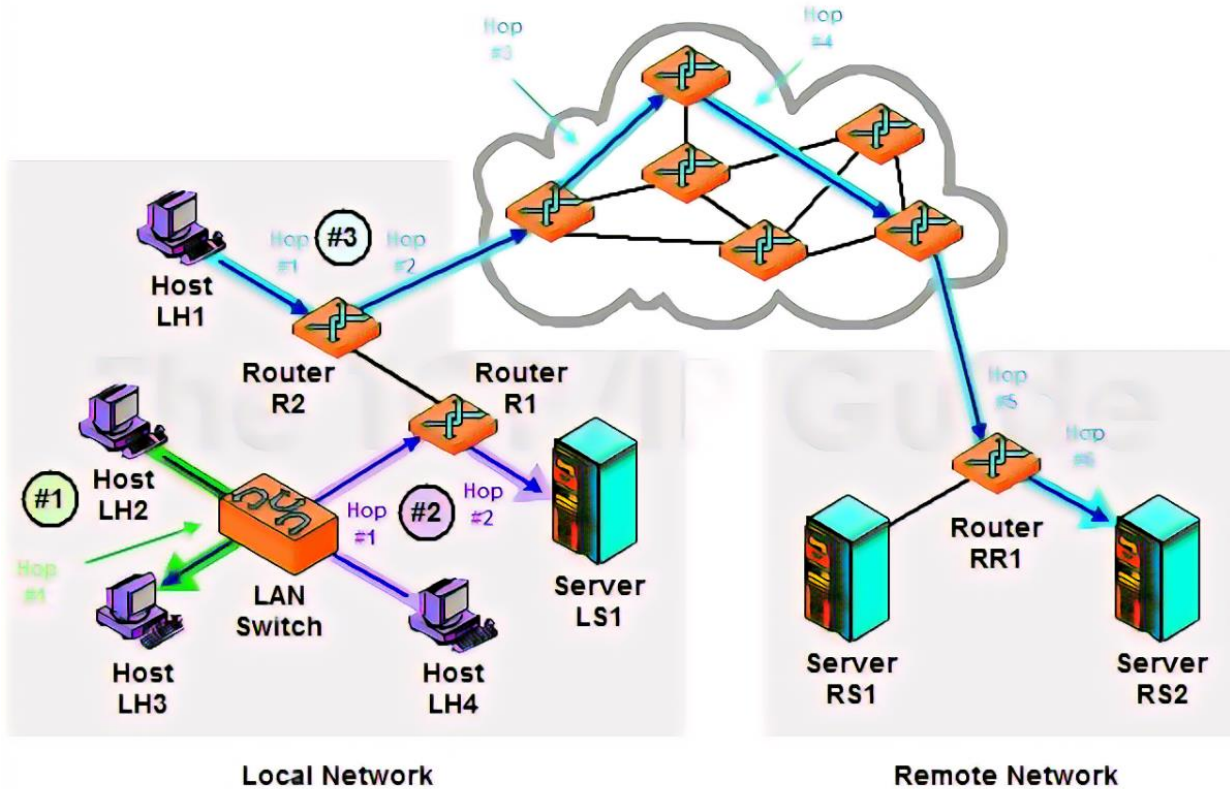


Http is **NOT** a protocol with encryption



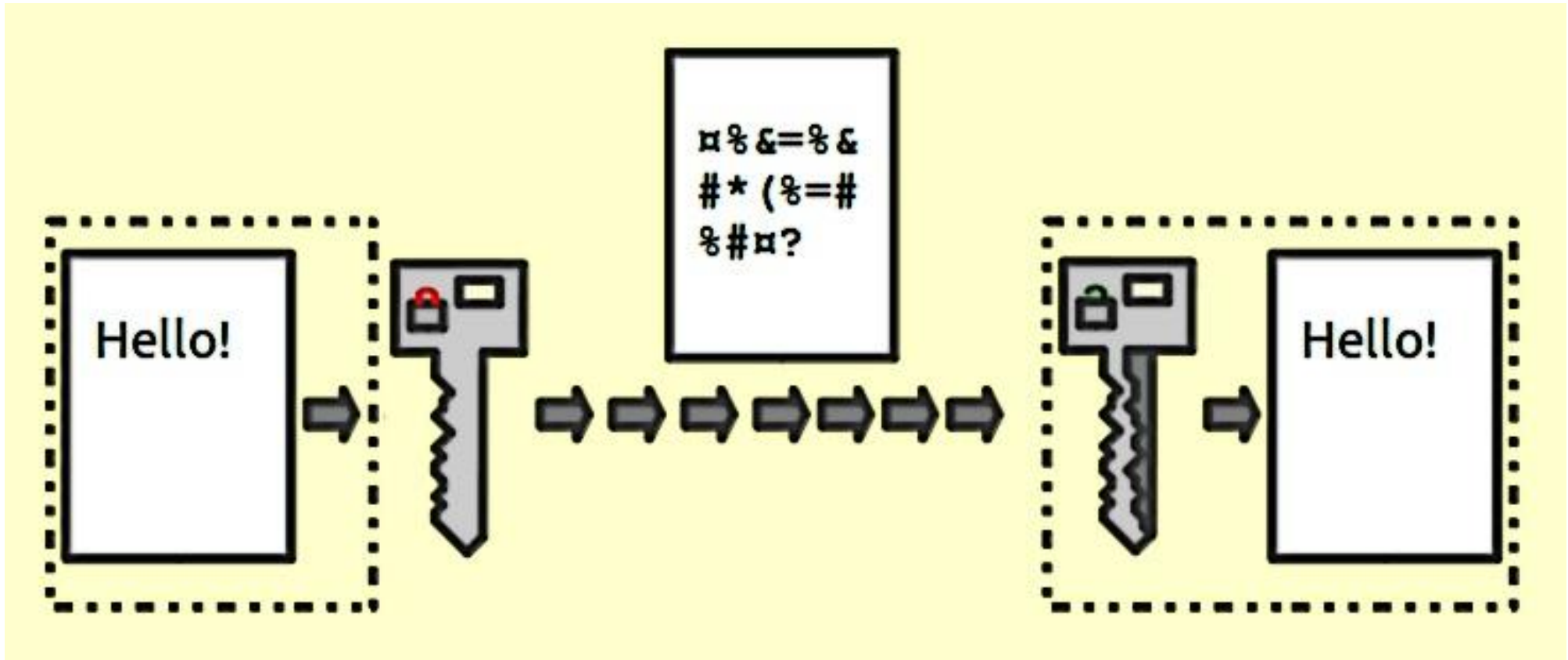
- All message in plain text.
- The biggest Vulnerability is Http protocol **itself!**

All middle nodes can **SEE** your http packs



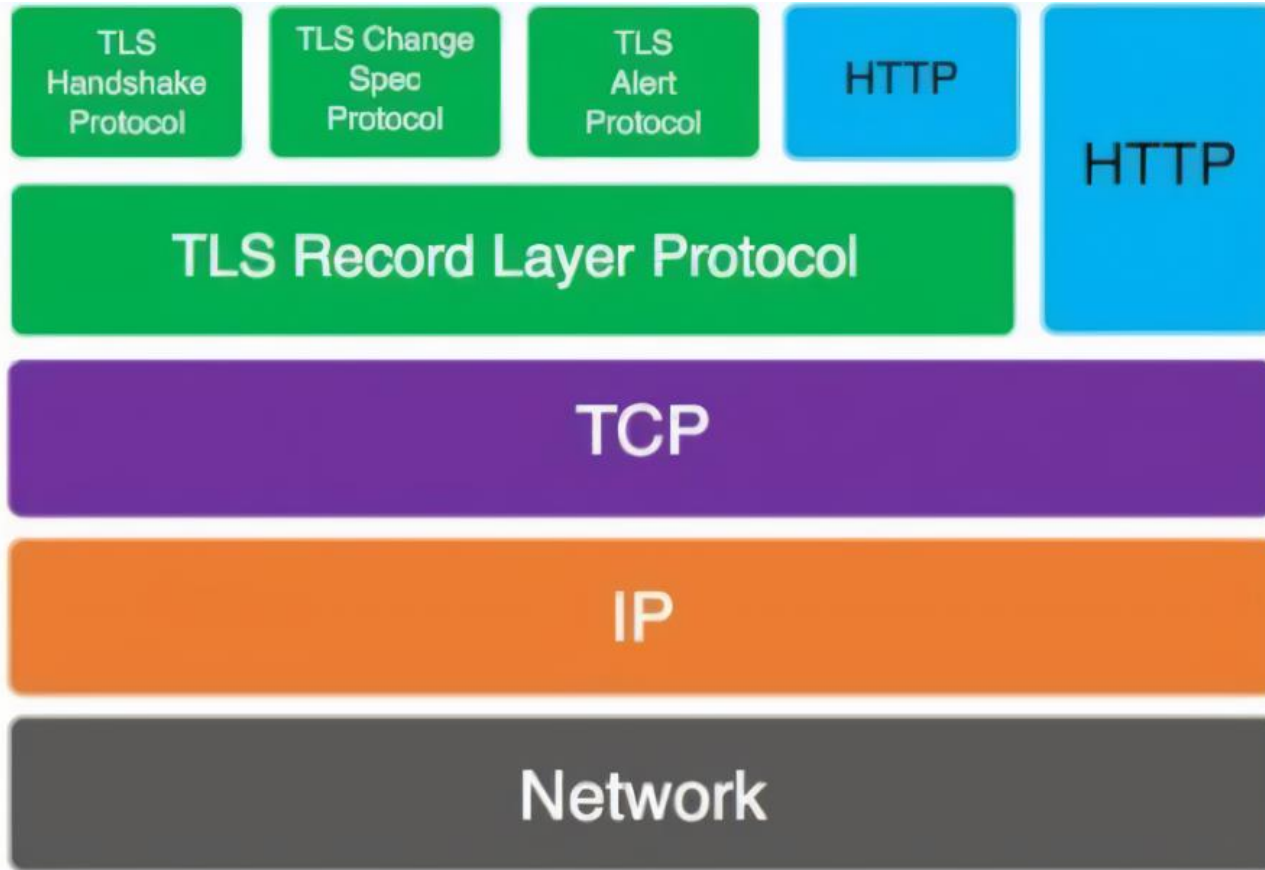
- TCP/IP packets pass through many nodes, and each node can see the entire content of the packets

Encryption with a Key (secret)



- Normally the algorithm is open and public.
- Without the key, the ciphered text cannot be decrypt.

Https - add a lock for the Http protocol



<https://v.qq.com/x/page/b0188l0shhr.html>

<https://v.qq.com/x/page/g0188shm2gf.html>

<https://www.zhihu.com/question/41432680>

Cryptography

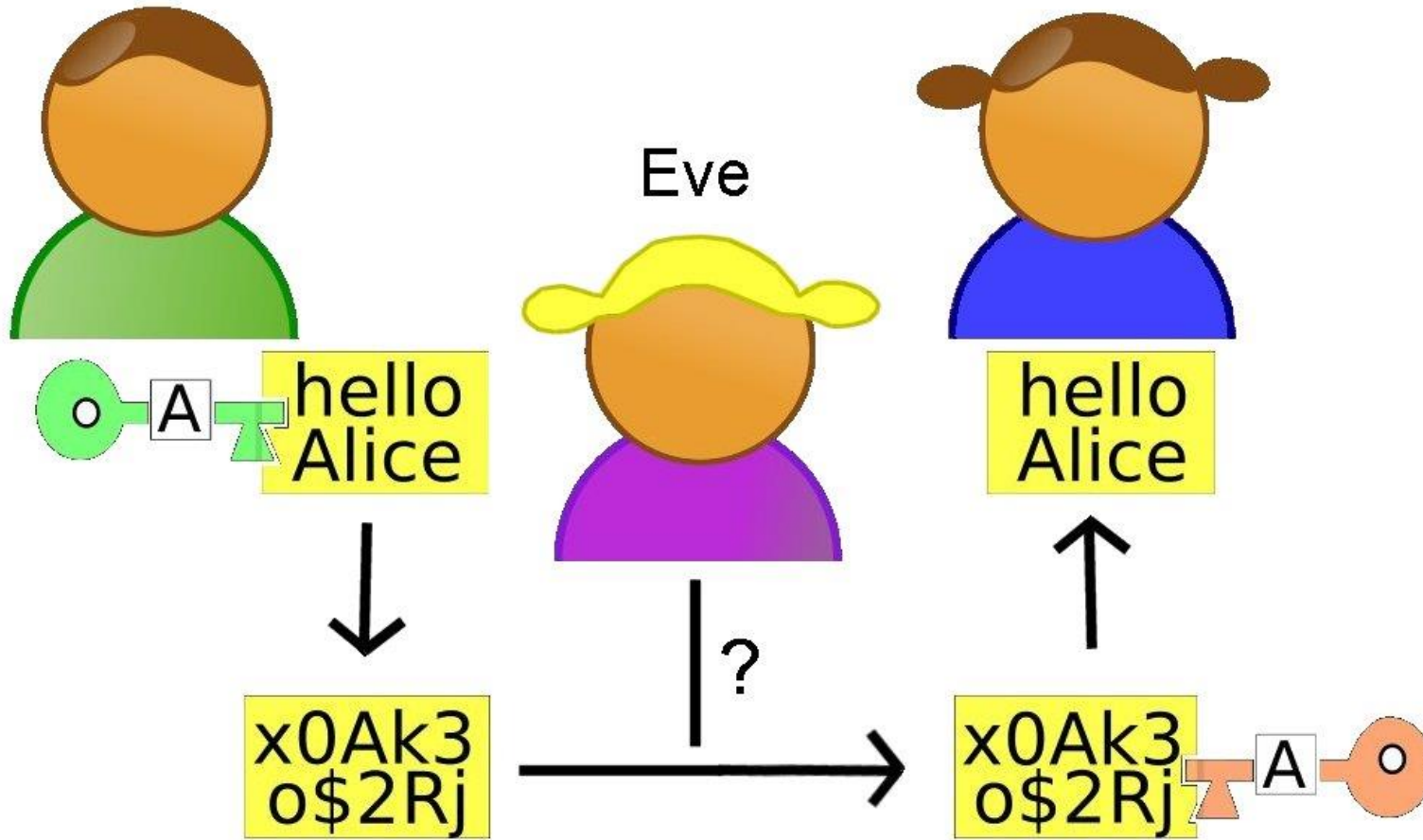
- Encryption/Decryption

- Hash functions

- Key exchange

- Digital signature

Bob, Alice and Eve



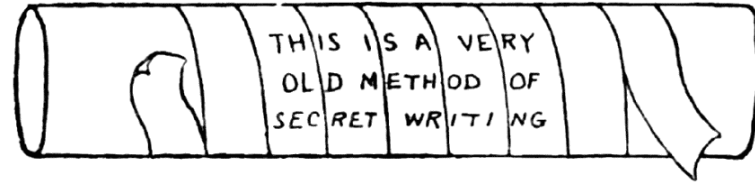
- The famous Eve!

Ancient encryption/decryption

芦花丛中一扁舟，
俊杰俄从此地游。
义士若能知此理，
反躬难逃可无忧。

--水浒传

TH OL SEC IS I D M RET S A ETH WR VE OD I TI RY OF NG



- The **Scytale** Cipher was used in ancient Greece by the Spartans in which a band was wrapped around a rod, and a message was written.
- You may try different rods to make it readable.

<https://www.forthright.com/a-guide-to-encryption-history-how-it-works-and-why-you-need-it/>

Dialect **is** an excellent cipher!

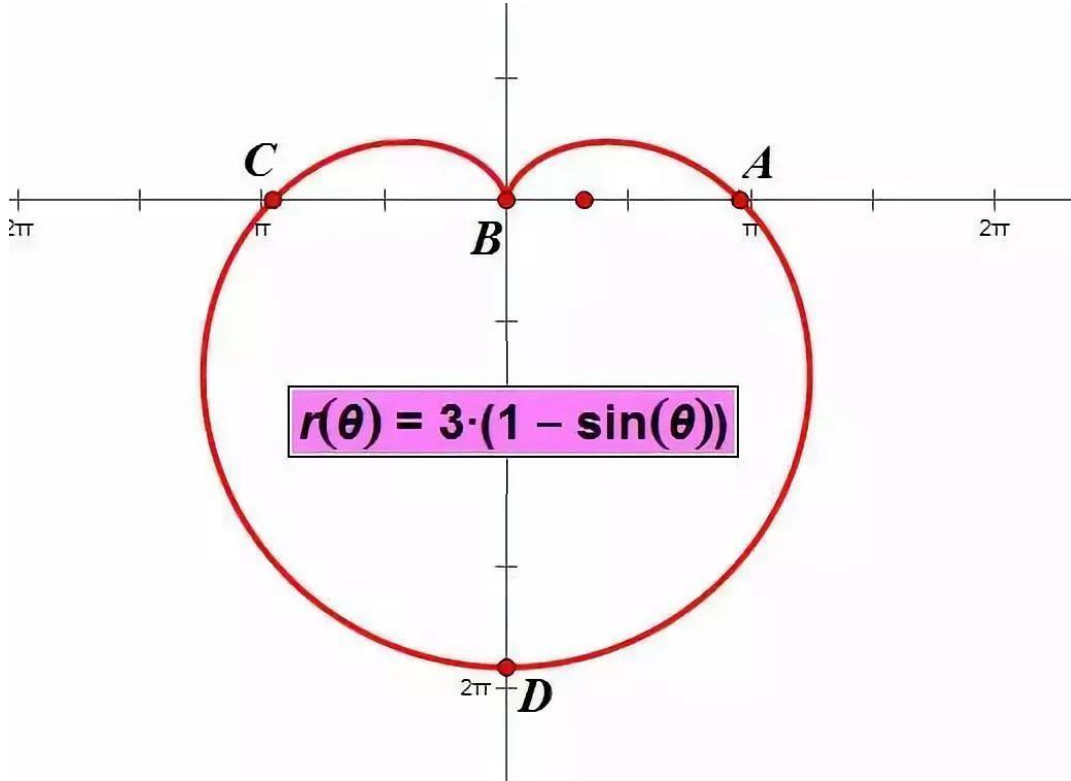


“Wind Talkers”

<https://www.zhihu.com/question/356444538>

<https://baijiahao.baidu.com/s?id=1583124320725288717&wfr=spider&for=pc>

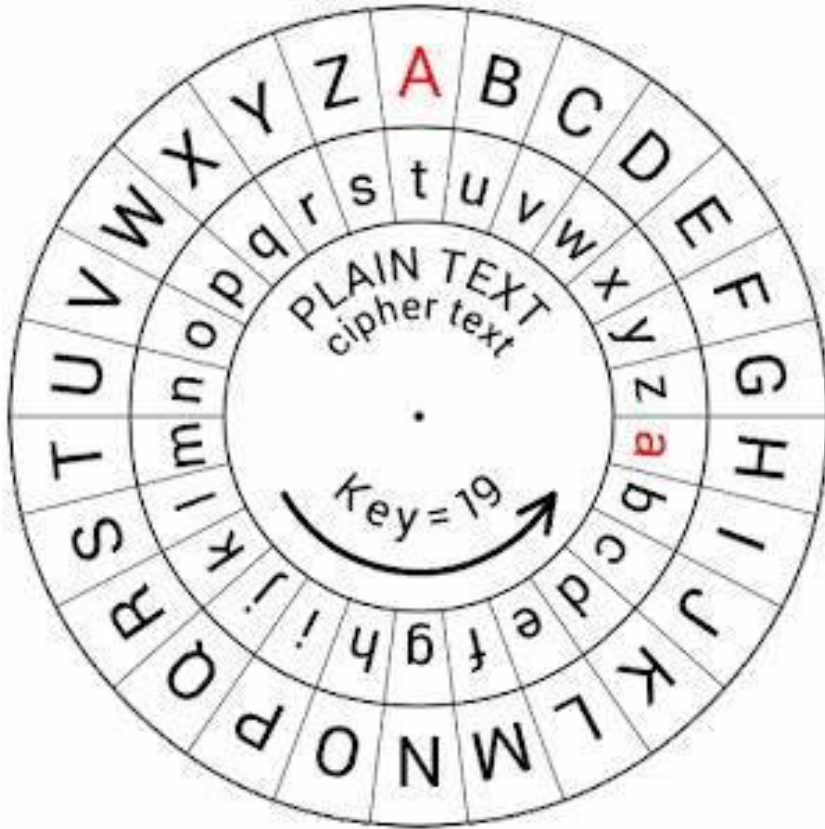
Encryption



Encryption is the process that scrambles **readable** text so it can only be read by the person who has the secret code, or decryption key.

While **Eve** always want to know the secret!

Caesar cipher



Encryption: $E(x) = (x + k) \pmod{26}$

Decryption: $E(x) = (x - k) \pmod{26}$

Crack the encryption

```
53*+305))6*;4826)4+.4+);806*;48+8'60))85;]8*:*8+83(88)5
*+;46(;88*96*?;8)*+(-;485);5*+2:*+(-;4956*2(5*-
4)8'8*;4069285);)6+8)4+;1(+9;48081;8:8+1;48+85;4)485+52
8806*81(+9;48;(88;4(+?34;48)4+;161;:188;+?;
```

数字8出现的频率最高，有34次，而英文中出现频率最高的字母是 e

8 => e

英文中最常见的单词是 the，而 ;48 出现了7次，很可能就是 the，可以得出

; => t
4 => h

; (88 很可能表示 tree

.....
继续分析

a good glass in the bishops hostel in the devils seat,
forty - one degrees and thirteen minutes, north east
and by north, main branch seventh limb east side,
shoot from the left eye of the deaths head, a beeline
from the tree through the shot fifty feet out.

字母 ↕	英语中出现的频率	
e	12.702%	
t	9.056%	
a	8.167%	
o	7.507%	
i	6.966%	
n	6.749%	
s	6.327%	
h	6.094%	
r	5.987%	
d	4.253%	
l	4.025%	

The Vigenère cipher

⊠	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Plain

tellhimaboutme

Key

cafecafecafeca

Cipher

veqppjiredozxoe

<https://www.bilibili.com/video/av71024212/>

<https://www.geeksforgeeks.org/vigenere-cipher/>

Crack Vigenère

K I N G K I N G K I N G K I N G K I N G K I N G
t h e s u n a n d t h e m a n i n t h e m o o n
D P R Y E V N T B U K W I A O X B U K W W B T

- **Repeatable results can be used as a weak**

https://www.sohu.com/a/467643712_121124372

Run Vigenère for many rounds

⧫	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

The Vigenère cipher table

Key1: Hellomycat

Key2: Dogissleep

Key3: Idislikemosquito

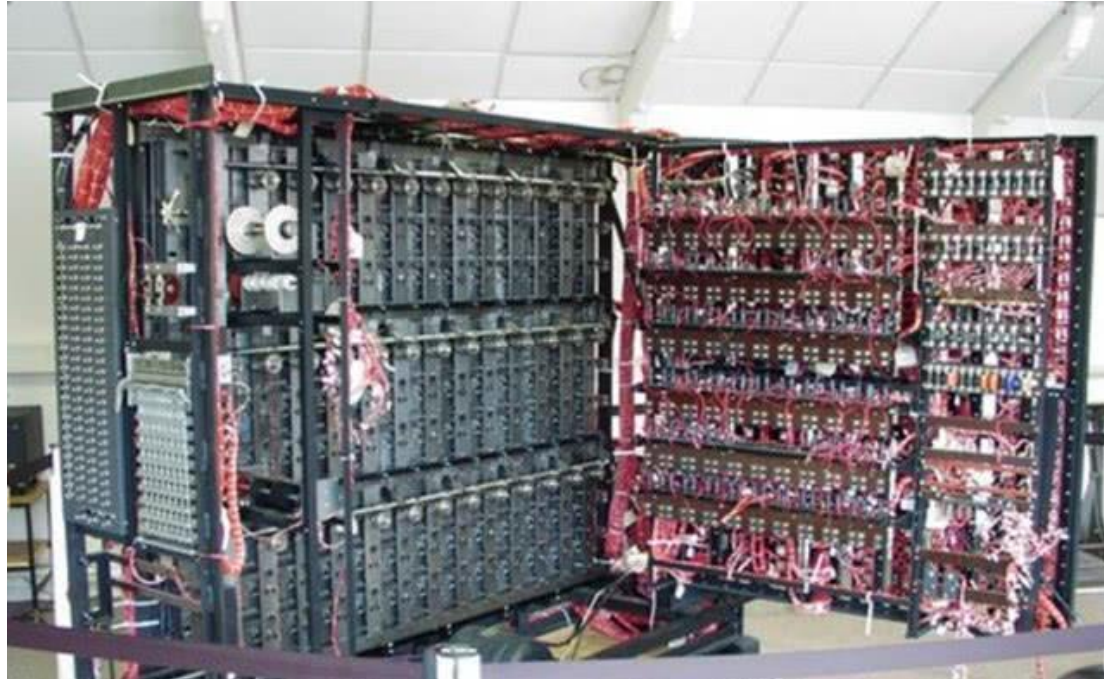
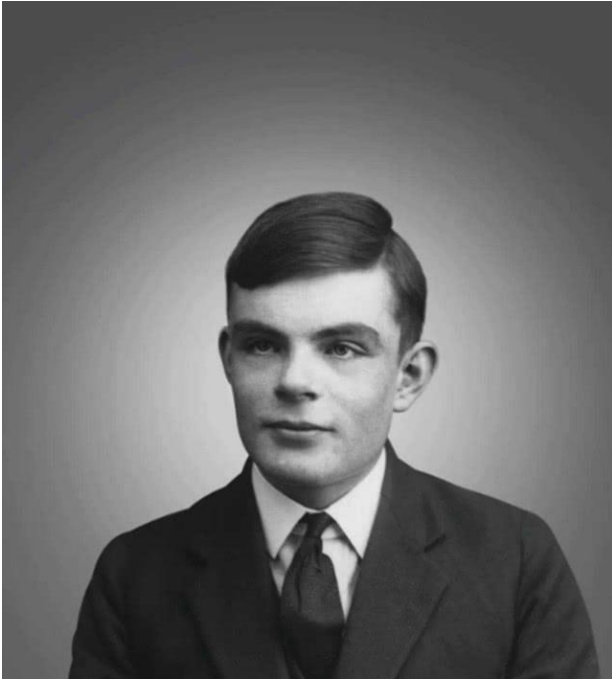
- You may need to do the encryption/decryption with longer time.
- It is difficult to manage so many keys.

Enigma machine



Using vector of wheels to generate many keys automatically

Security is always: "The code"



Alan Mathison **Turing**, 1912 - 1954

https://www.sohu.com/a/443659675_120339167

<https://www.163.com/dy/article/EHTEK00805372PI2.html>

<https://www.163.com/dy/article/GDC5UVCC0543U41J.html>

<https://item.jd.com/12307986.html>



Crack the Egyptian Script

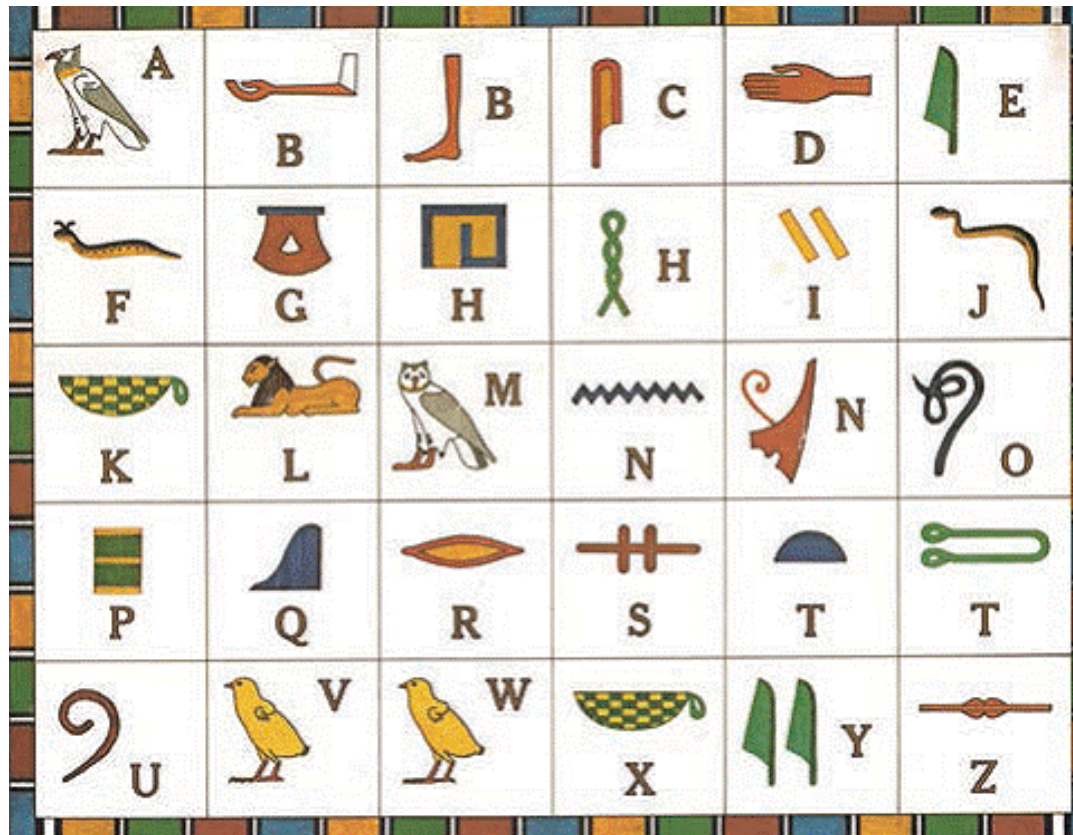
第一部分: 古埃及象形文字
Egyptian Hieroglyphics Script

第二部分: 古埃及草书
Egyptian Demotic Script

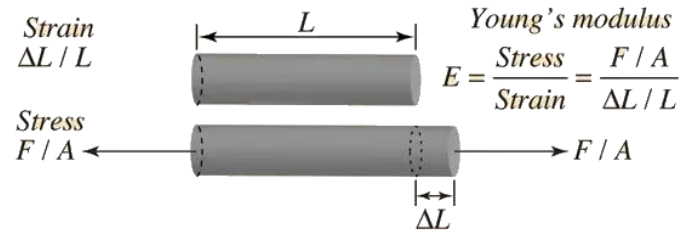
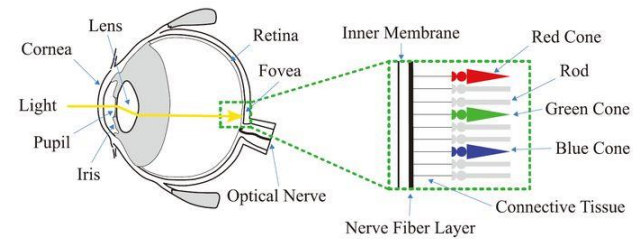
第三部分: 古希腊文
Greek Script



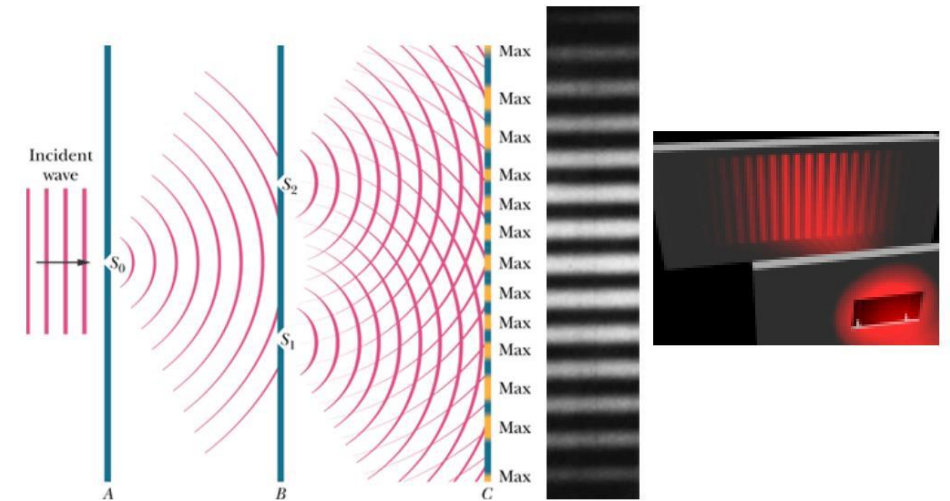
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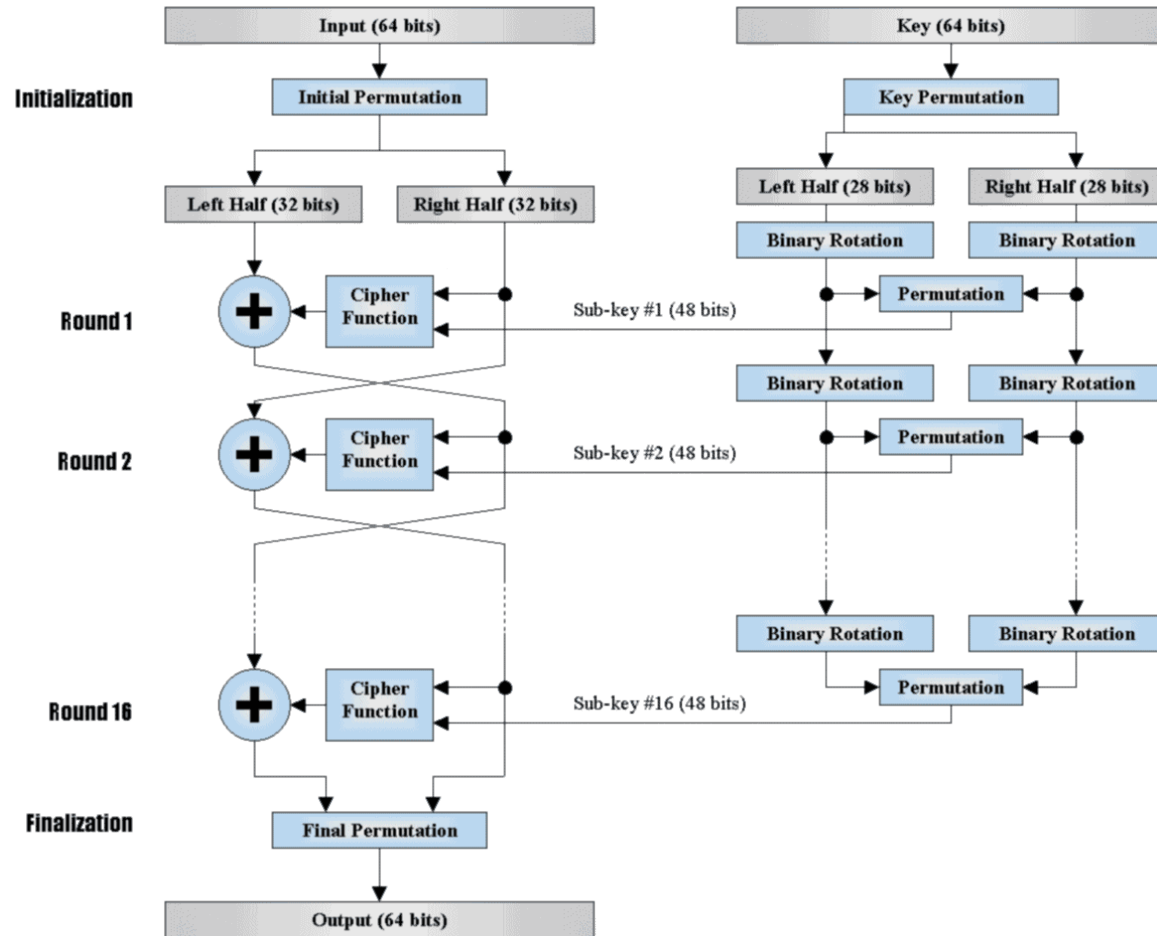
Crack the Egyptian Script



Young's double slit experiment



DES (Data Encryption Standard)

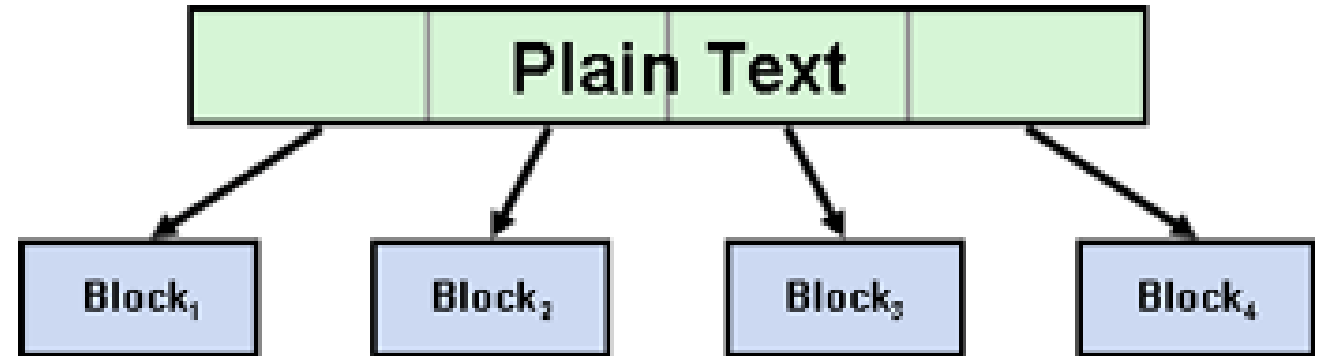


- In the early 1970's, IBM realized that their customers were demanding some form of encryption, so they formed a "crypto group" headed by Horst-Feistel. They designed a cipher called Lucifer.
- In 1973, the Nation Bureau of Standards (now called NIST) in the US put out a request for proposals for a block cipher which would become a national standard.

<https://www.bilibili.com/video/BV1KQ4y127AT>

<https://www.cnblogs.com/songwenlong/p/5944139.html>

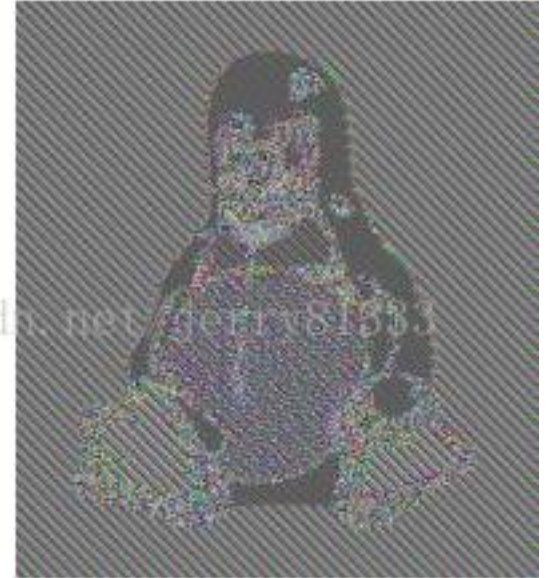
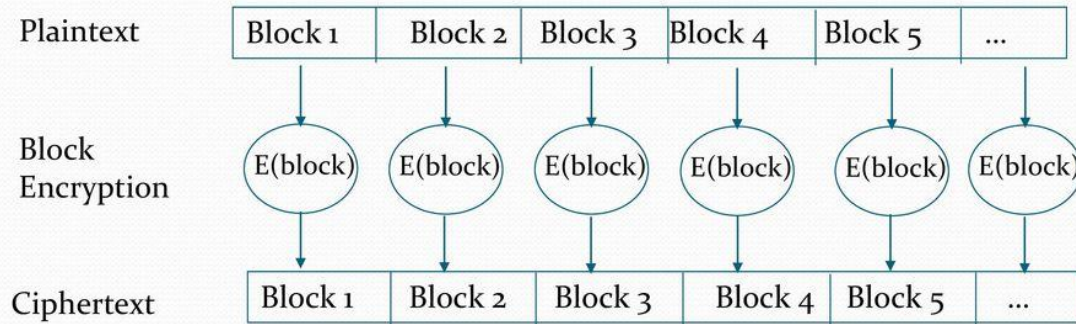
Block cipher



- Normally the data length is long than the key.
- The data need to be put in blocks

Block cipher modes

Electronic Code Book (ECB) Mode



- **ECB** is most simple one but sometime it is not good enough.
- There are many other modes like **CBC**、**PCBC**、**CFB**、**OFB**、**CTR**
- No simply good or bad between different modes, it depends on applications.

<https://blog.csdn.net/jerry81333/article/details/78336616>

Crack the DES by Brute Force



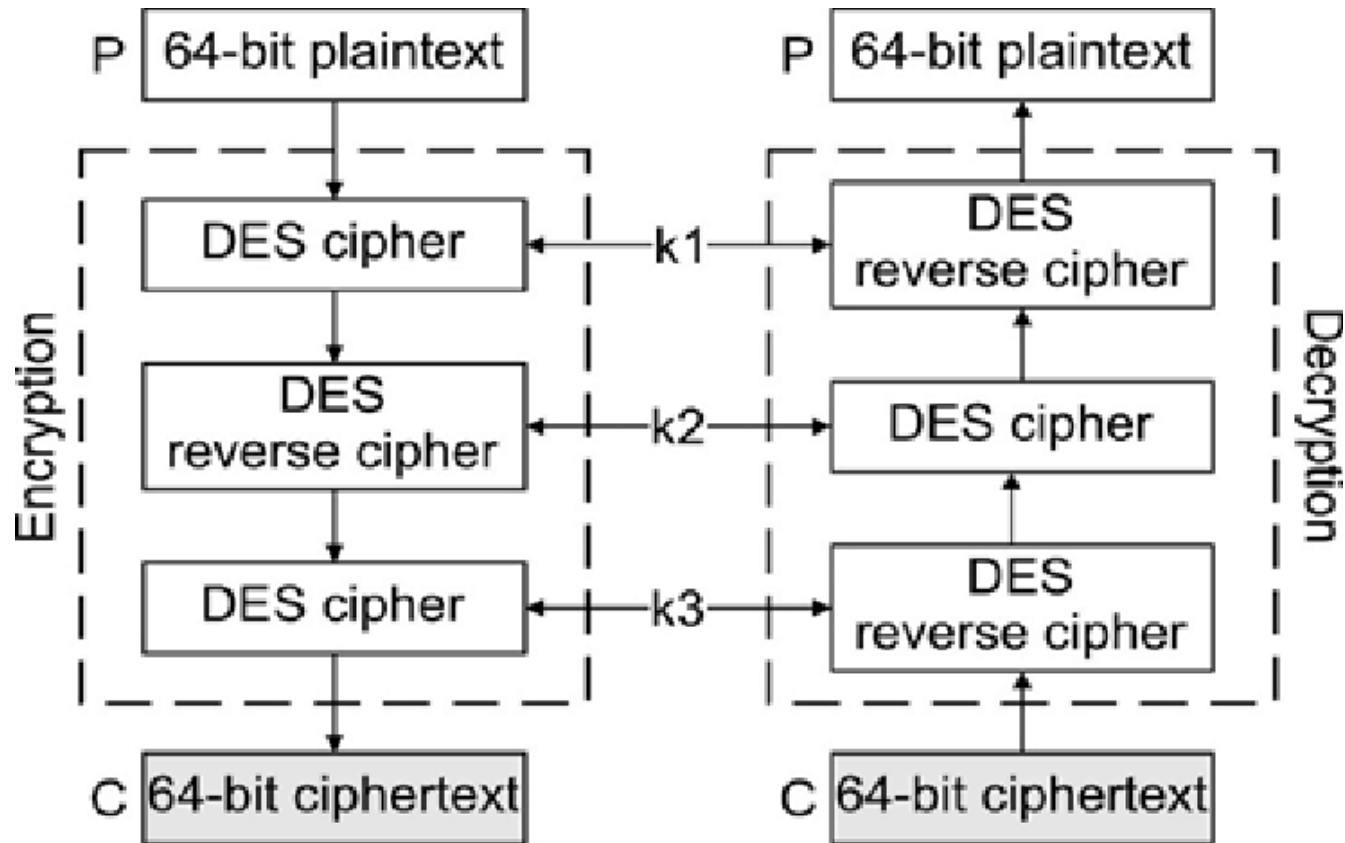
3gA8bjqI	m5MoKRJJ	0WZCBfwI	zm9jqUiN	glYqI53X	eokUQTMr	LDzoPv9e	qXUCvXrT	Enju6m2B
oiKq1eah	9WtsbkvW	W3gyedbV	12j9P5xy	8SxkUE9w	nXXADMm5	GiB1Z6l0	jio5wIbp	gozIsTIm
lzOWmgPt	i9XnarBa	yo0acST0	Q1acFGJq	IZkFcv0u	5ugNFv1w	w2G17xIN	w2bejXwY	RvYxtMjU
nRl06HW5	chRSQrAb	DoqQQrgd	RTUevfn2	faFLIMRf	Mm1nJoZE	JilIRGJ5	crEHftjc	eDmS5WJp
RfVugGsl	RgsGp0oU	tHd41yuV	L0x6wZkG	IdqZQtXK	TIBeHAVa	XYVA4UQI	VXoncM5C	uk3Eaauk
dB90qMas	DJrU178v	nCZrMpk8	Xaup9lw5	miufIFTB	Z9p5K9Ut	suAjQYep	x5CosAtw	bHTdQPkj
A1gr9Utx	Pik7Il04	vaYLGHjc	BtJ8ktAk	au0greB1	P9FTnI7c	a6UEr0cr	iEp0z3tC	Hzg7iYWZ
6FFcHAoe	Yfa3SY5I	351sV8w5	J3PxPYNz	WGjLGUBW	c2503M6c	pDfsu2Q4	cdP5cwB5	9vFjEHQu
2MxkJa3i	B4bLGWH4	UIJcx0ns	IMT1fNa3	PASSWORD	mfviEj5x	EsKPneug	GKJU0utG	FK92JFQ3
rPlowpJr	Yr30oFJ5	GHcDJqvX	A3QA5Ye3	YbtwXwnn	NGJLCNL8	2vJsptvH	zCinx0EC	UN3j3pXC
vmjRD4i0	Q1kh5j6Y	5i6TSEaT	lId407YG	deYv90Sn	2nczWHh6	vFXjiFRI	4sDHxCZm	Qpe5zL30
4eggPjtZ	KRfuFRnU	VtQhZlv9	XV9DkP4x	S9mMEd5S	bXyfJTGK	NQxNST0H	qfSCnY1M	WjJz8X2c
9rpYjpuU	ZS69eKWL	7iMwKrlo	mtCQSeYd	mmam9dn9	5ha4ddzy	o9KYUF5Y	fJAzwIdn	zzHoKGY1
DDGTFjZL	Yt7Fm3lv	zqJ8pdW1	7YcJfnB9	5oywq9fK	3sA0ewmA	zHJyTRNe	UupQ0TkY	XJpvqp2B
q3LW0tcJ	4Pm6aoip	iE9NzJgc	ntSxFnxl	qW7eAqKE	1VmD6lf0	5uzTxti6	2gRsURBz	yxseYGgg
beCDYzD2	dcrV4A54	jaT6KoQH	MtEulhJ3	xt67N62g	zKQIfxdI	KbBFpDhY	Qd5PGAPW	csfwIRrf
UAid0Mw8	ZDqQ2xZq	QIJfG6Se	prhomHT2	scLAHNAS	NmIQ0UFQ	7TqPhSZP	kp7MUZMk	WSKd1TOU

- Try all possibility of the key
- DES is not safe in current world

<https://www.bilibili.com/video/BV1KQ4y127AT>

<https://www.cnblogs.com/songwenlong/p/5944139.html>

Replace DES with 3DES

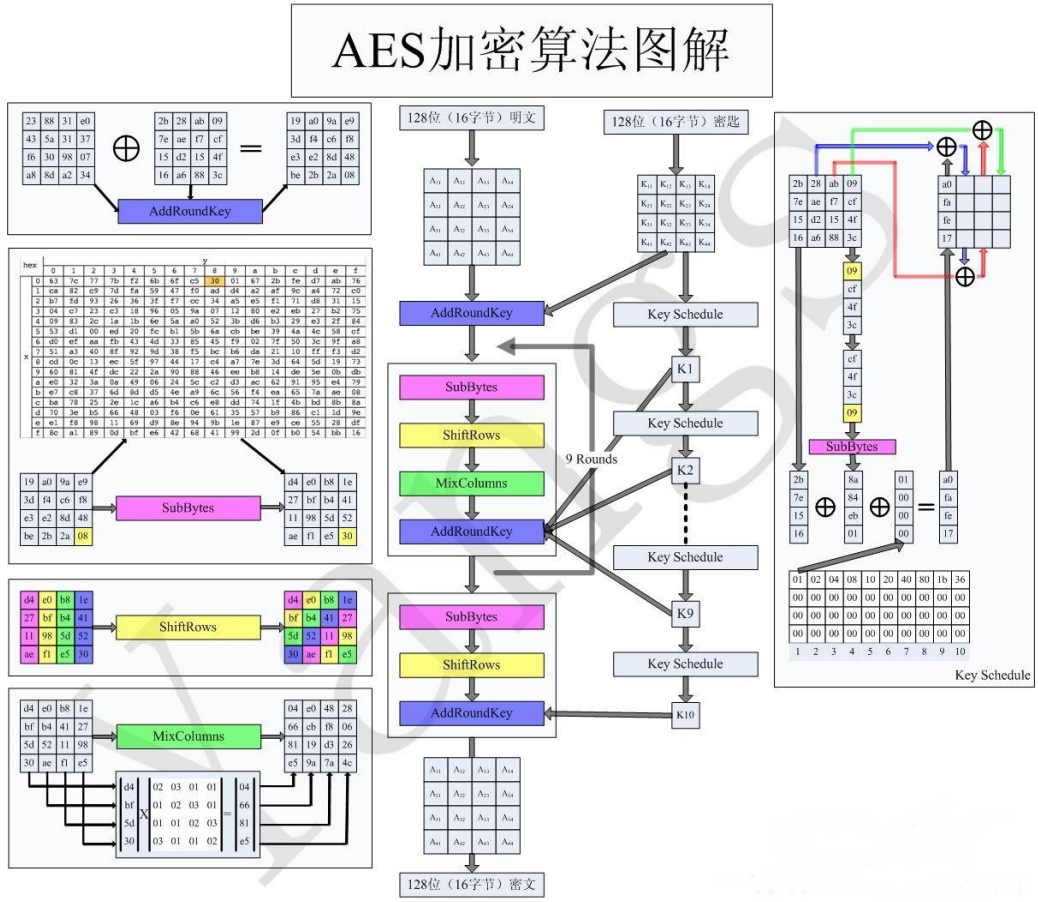
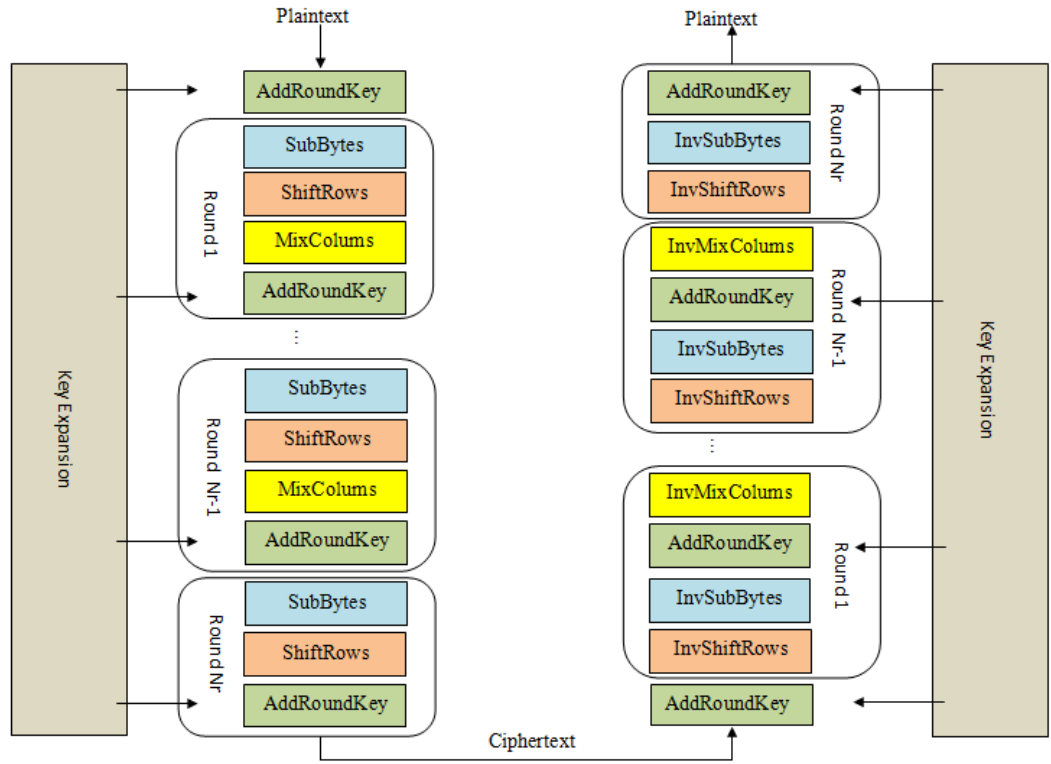


- In 1997, DES is not regarded as safe one due to the length of key.
- So do it triple times and got 3DES.

<https://www.bilibili.com/video/BV1KQ4y127AT>

<https://www.cnblogs.com/songwenlong/p/5944139.html>

AES (Advanced Encryption Standard)



Rijndael is a family of block ciphers developed by Belgian cryptographers Vincent Rijmen and Joen Daemen. It was submitted as an entry to the National Institute of Standards and Technology's (NIST) competition to select an Advanced Encryption Standard (AES) to replace Data Encryption Standard (DES).

<https://www.commonlounge.com/discussion/e32fdd267aaa4240a4464723bc74d0a5>
https://www.bilibili.com/video/BV1i341187fK/?spm_id_from=333.788.b_7265636f5f6c697374.2

Symmetric Encrypt/Decrypt algorithm

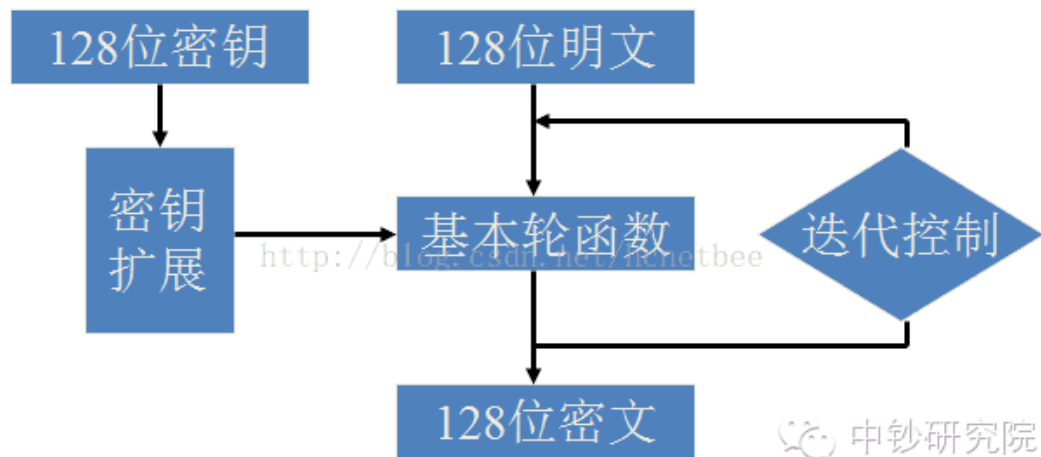
Algorithms	Key size	Block size	Round	Structure	Flexible	Features
DES	64 bits	64 bits	16	Feistel	No	Not structure, Enough
3DES	112 or 118 bits	64 bits	48	Feistel	Yes	Adequate security
AES	128,192,256 bits	128 bits	10,12,14	Substitution, Permutation	Yes	Replacement for DES, Excellent security
RC4	Variable	40-2048 bits	256	Feistel	Yes	Fast cipher in SSL
RC6	128-256 bits	128 bits	20	Feistel	Yes	Good security
BLOW FISH	32-448 bits	64 bits	16	Feistel	Yes	Excellent security

Chinese algorithms of symmetric encryption

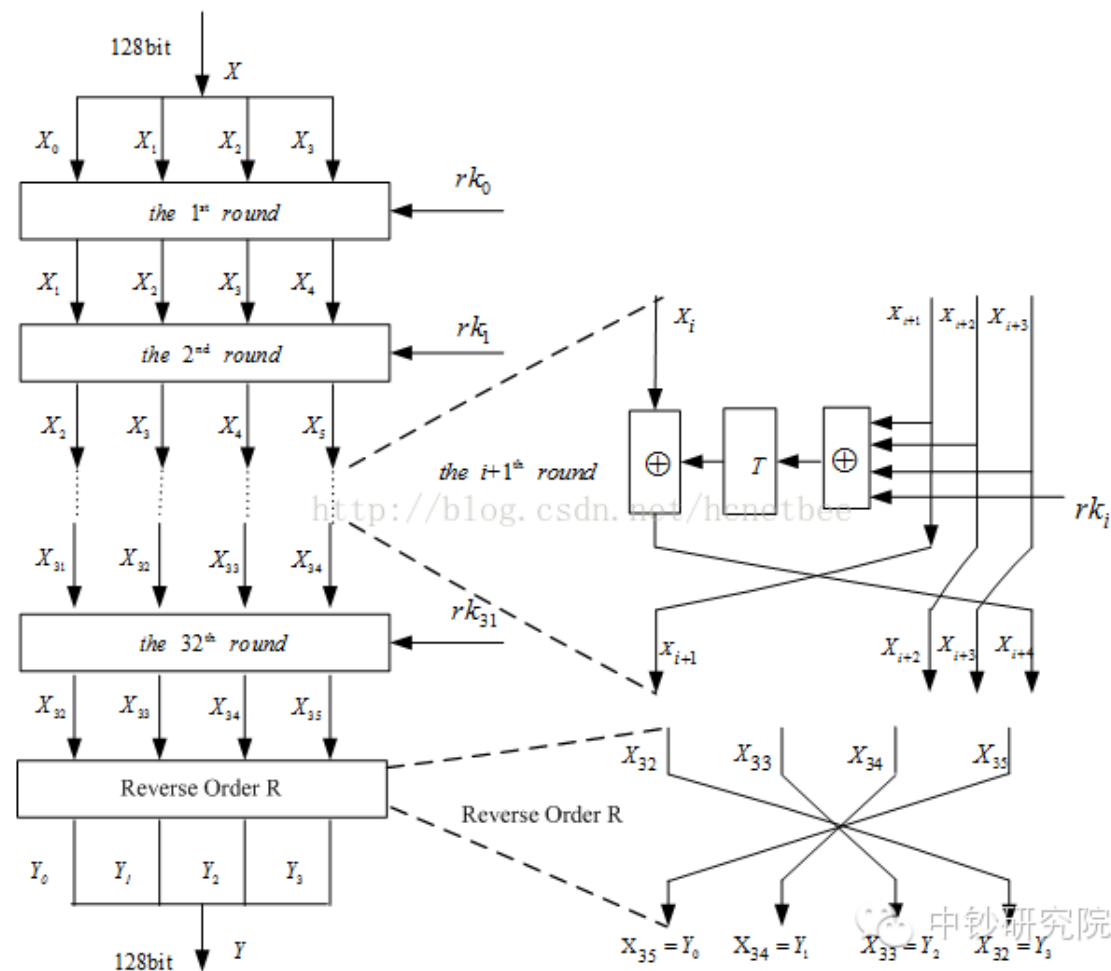


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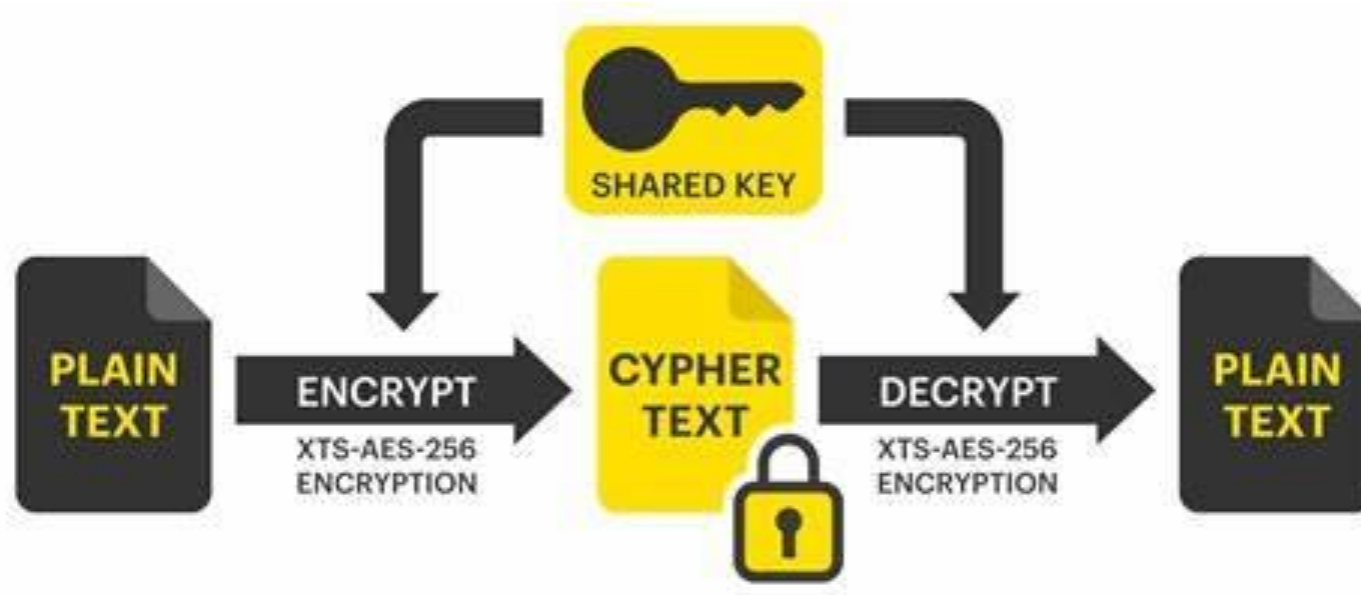
- SM1 is not open to public.
- SM4 is similar to AES.
- SM7 and ZUC are two new algorithms.



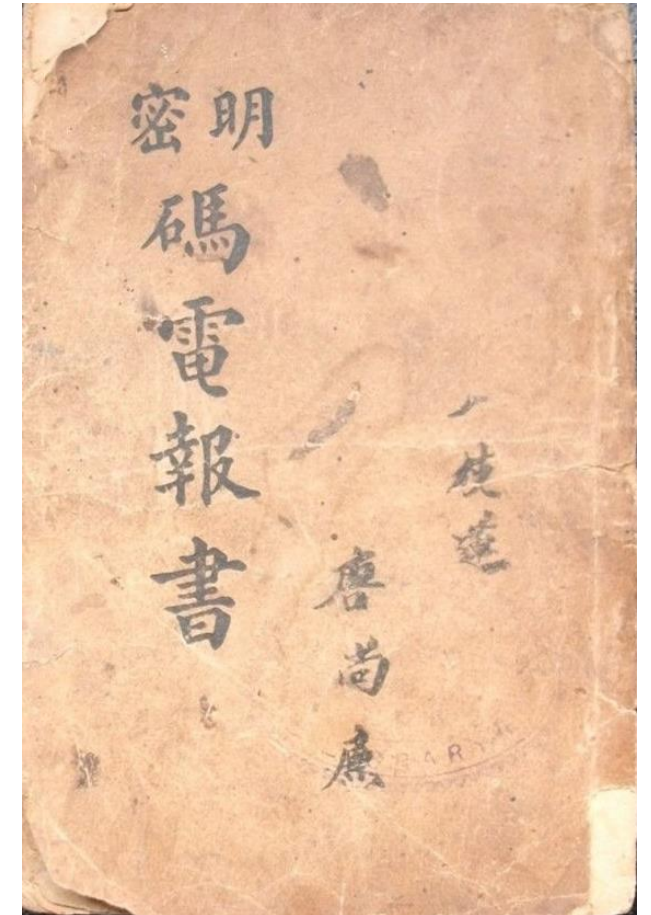
<https://blog.csdn.net/andylau00j/article/details/80102983>

https://www.sohu.com/a/288876376_100078137

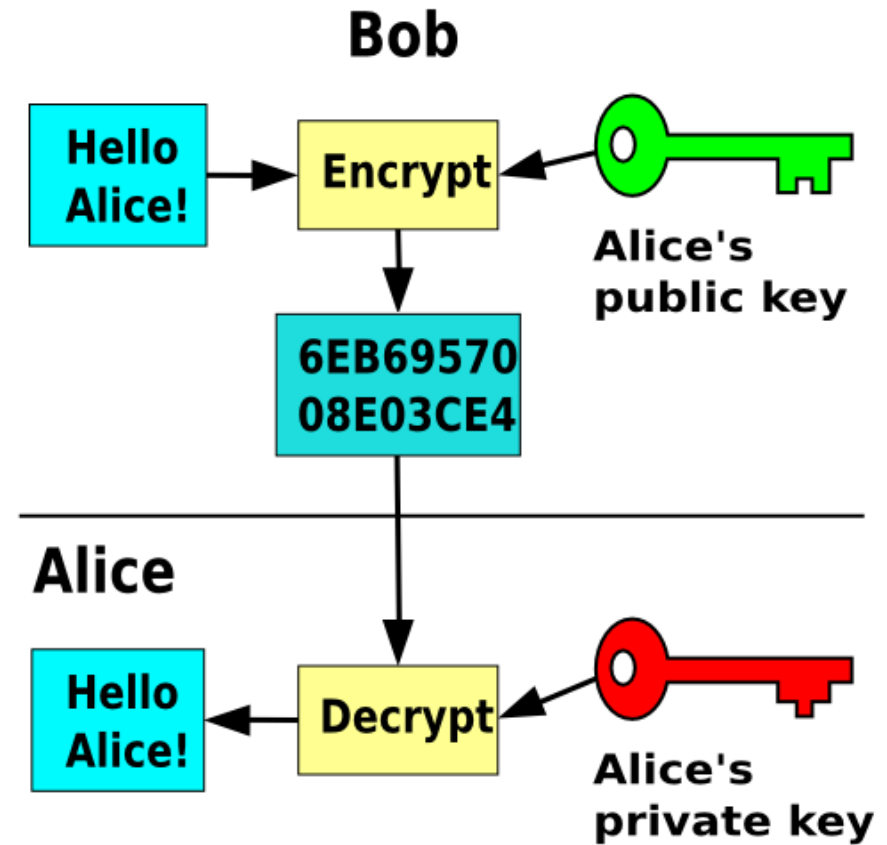
How to exchange the key safely



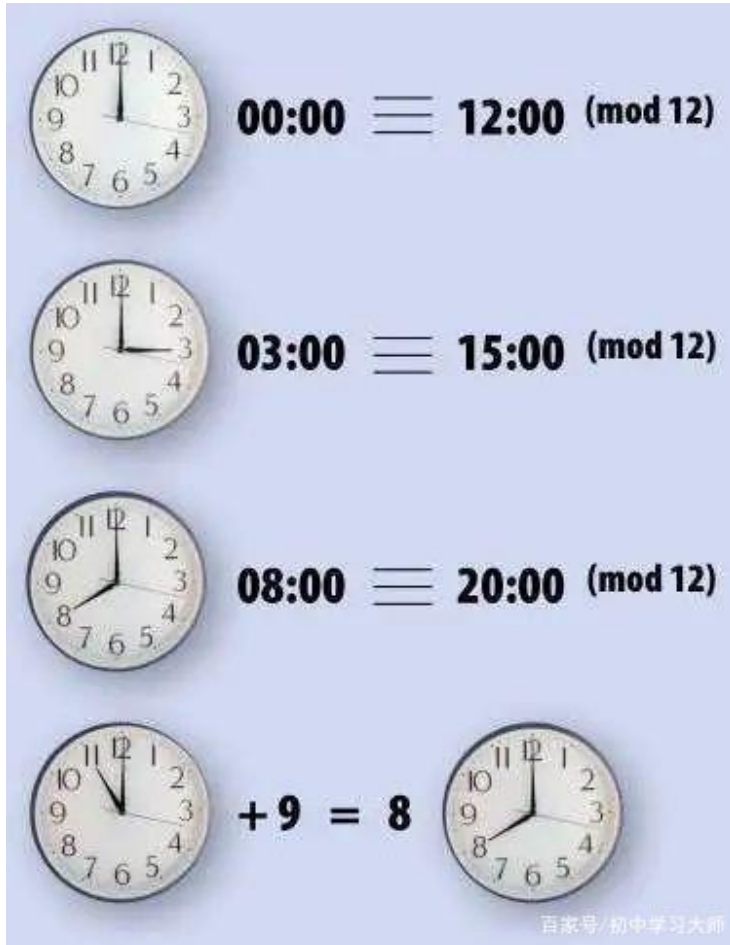
- Without a secure channel, the key cannot be exchanged safely!



Asymmetric algorithms



Math foundation: ^ and %



- 周而复始

Question: $a^b \% p$??equal or not?? $[(a \% p)^b] \% p$

Let $a = mp + q$, then $a \% p = q$

$$a^b = (mp + q)^b = C_b^0(mp)^b + C_b^1(mp)^{b-1}q + \dots + C_b^b q^b$$

1

Binomial expansion

All of them have p , so mod p would be 0

Then

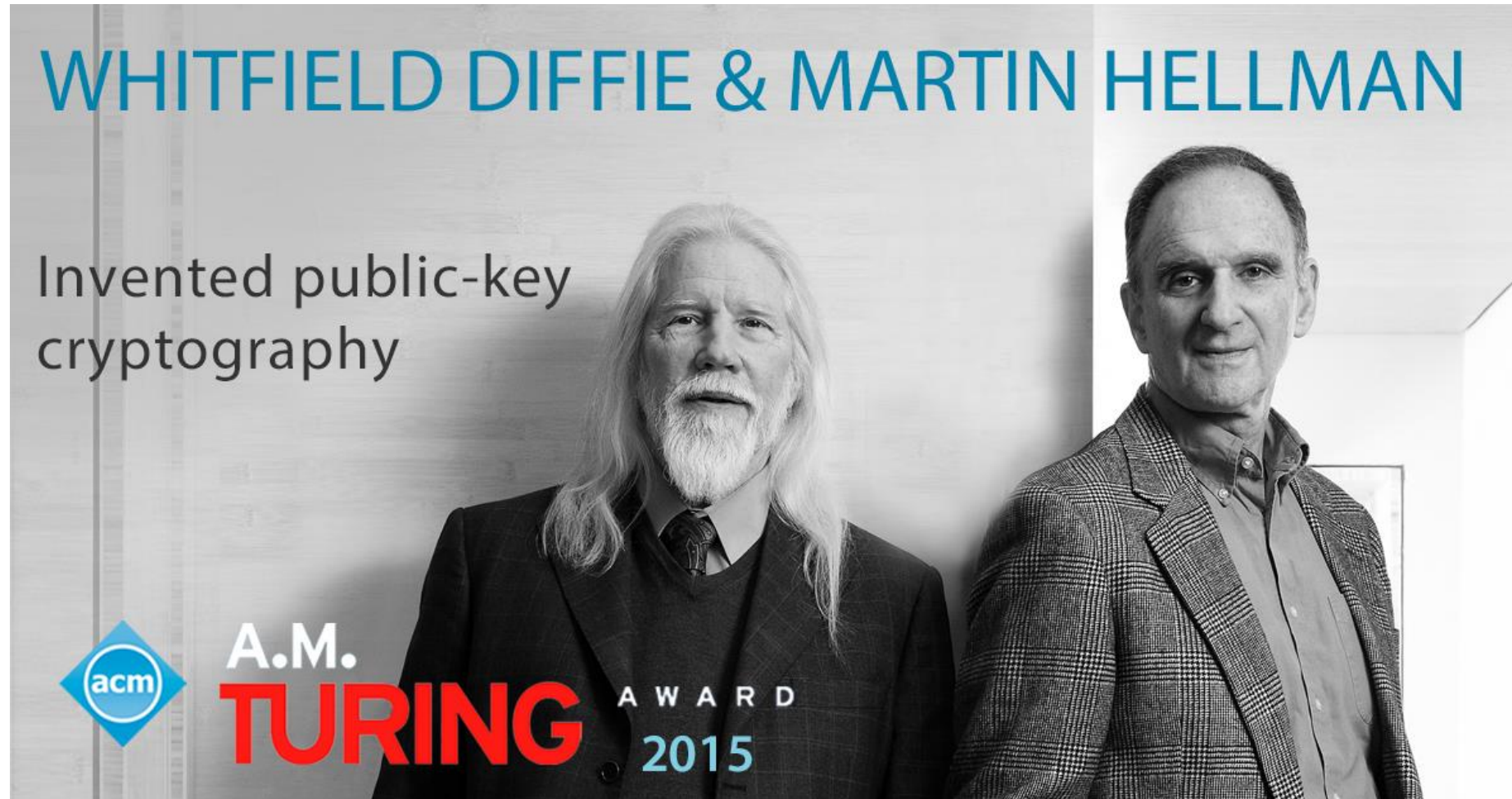
$$a^b \% p = (mp + q)^b \% p = q^b \% p$$

$$[(a \% p)^b] \% p = [(mp + q) \% p]^b \% p = q^b \% p$$

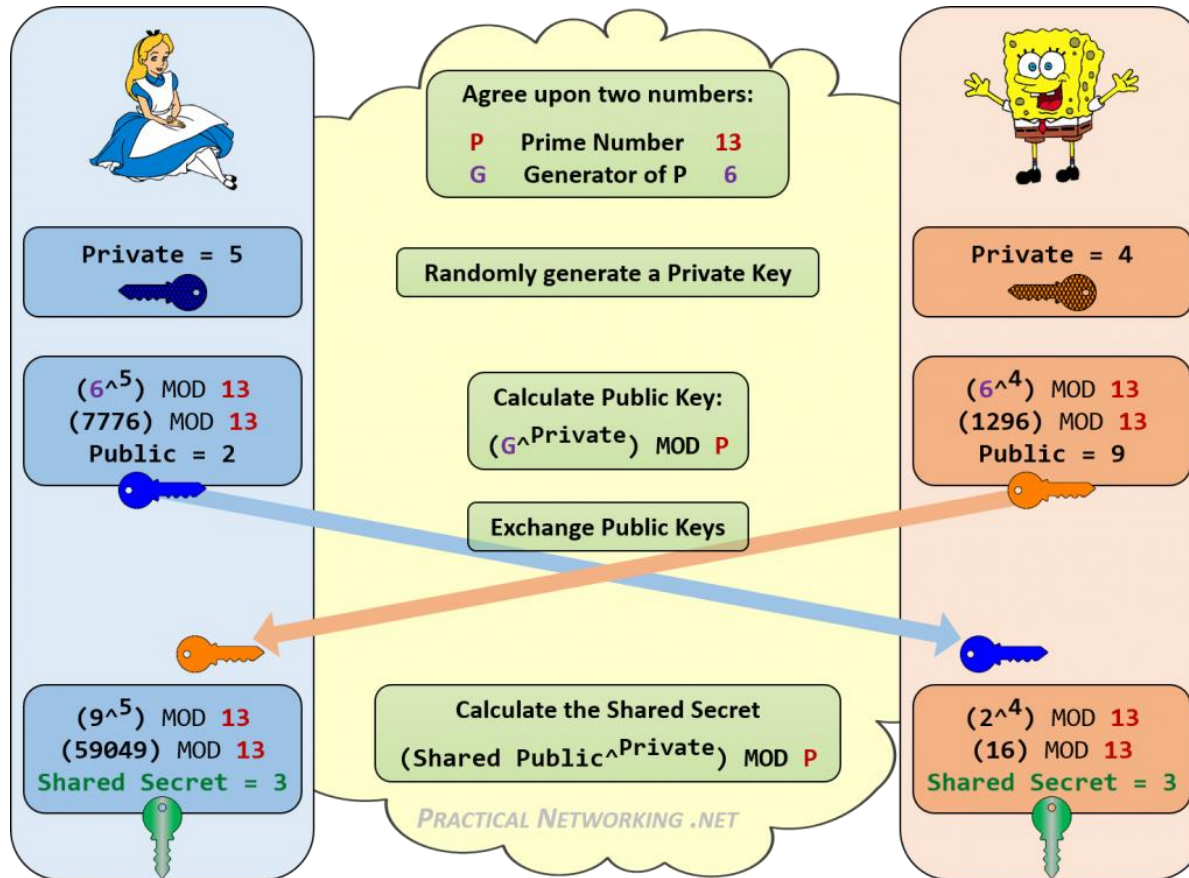
Answer: $a^b \% p$ equal $[(a \% p)^b] \% p$

- **Modular + Exponentiation algorithm**

Generate shared key by **A**symmetric algorithms



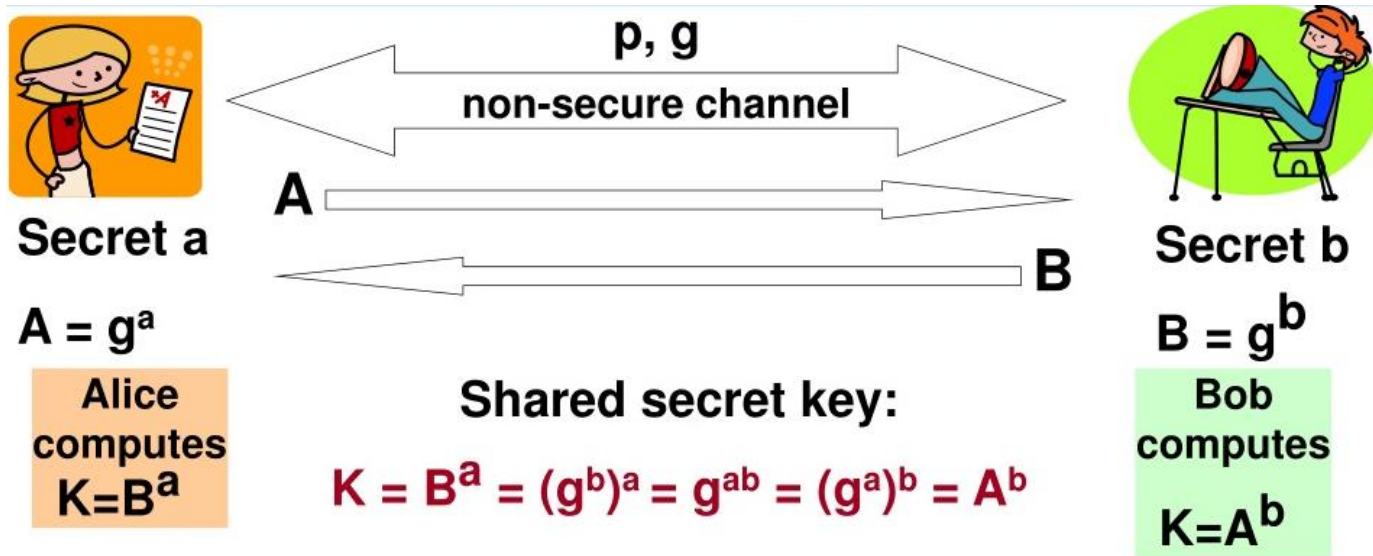
Diffie-Hellman algorithm



- p must be a primitive number
- g must be the **primitive root** of p

- **Public agree a pair of (p, g) .** Both parties choose their own private keys as a and b .
- Exchange **Public A** ($g^a \text{ mod } p$) and **Public B** ($g^b \text{ mod } p$)
- Session Key = $(\text{Public A})^b \text{ mod } p = (\text{Public B})^a \text{ mod } p = g^{ab} \text{ mod } p$

Crack the Diffie–Hellman algorithm

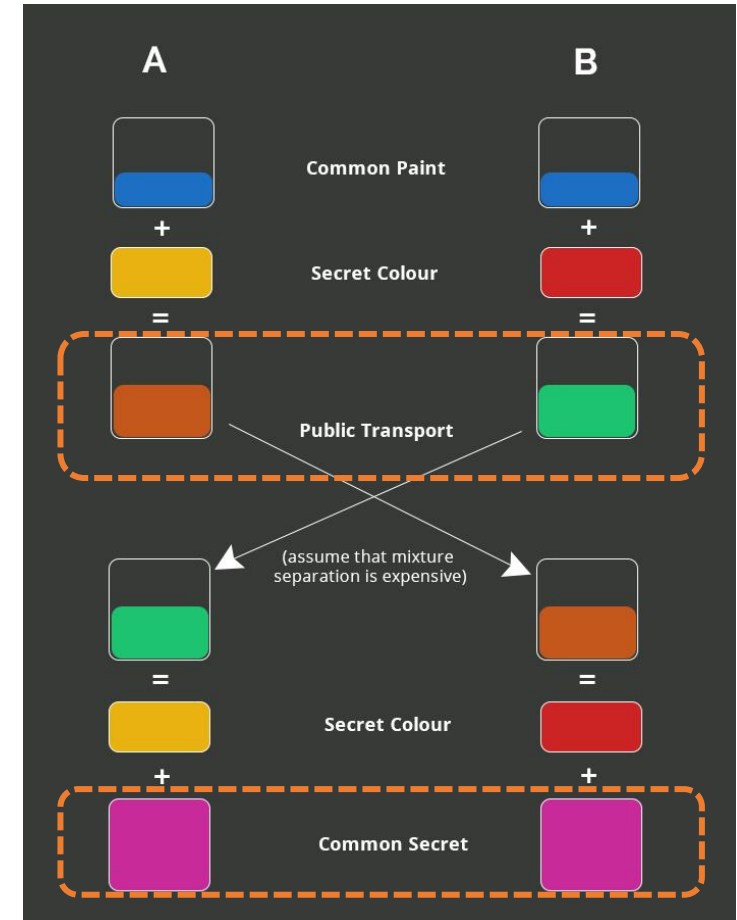


A possible way to find $K = g^{ab}$: knowing p, g, A , find a , such that

$$g^a = A \pmod p$$

which is the Discrete Logarithm Problem

Or a tough mission!

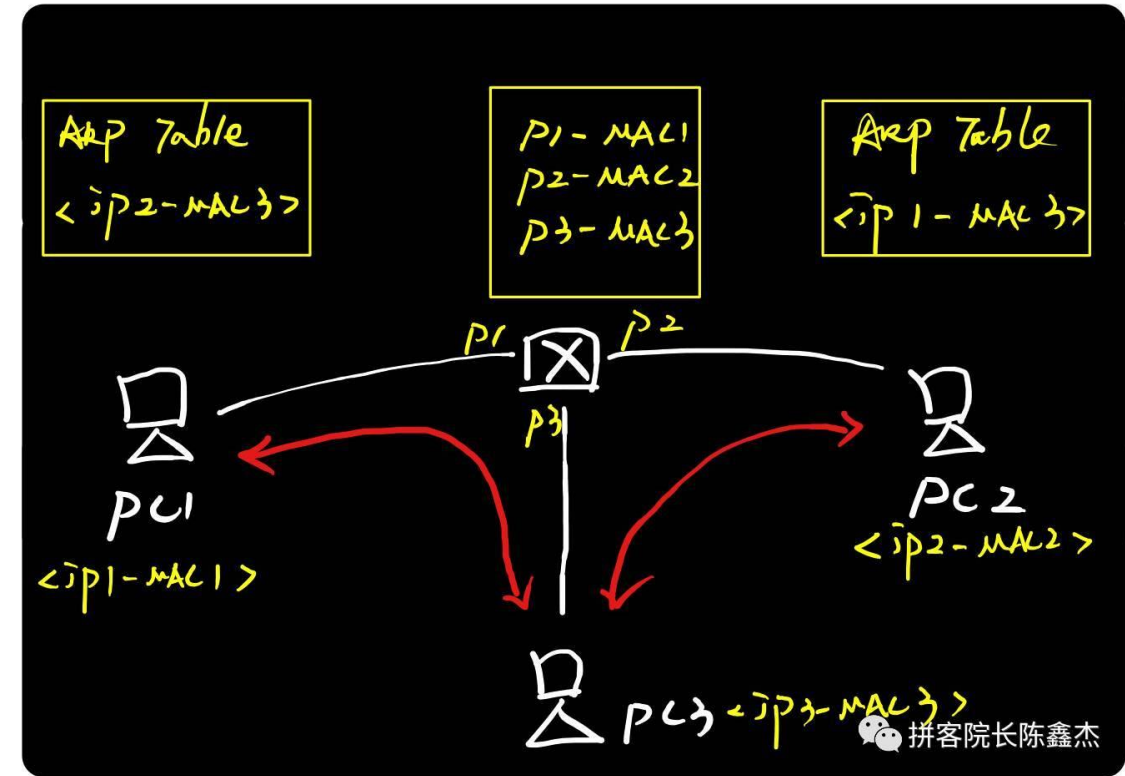
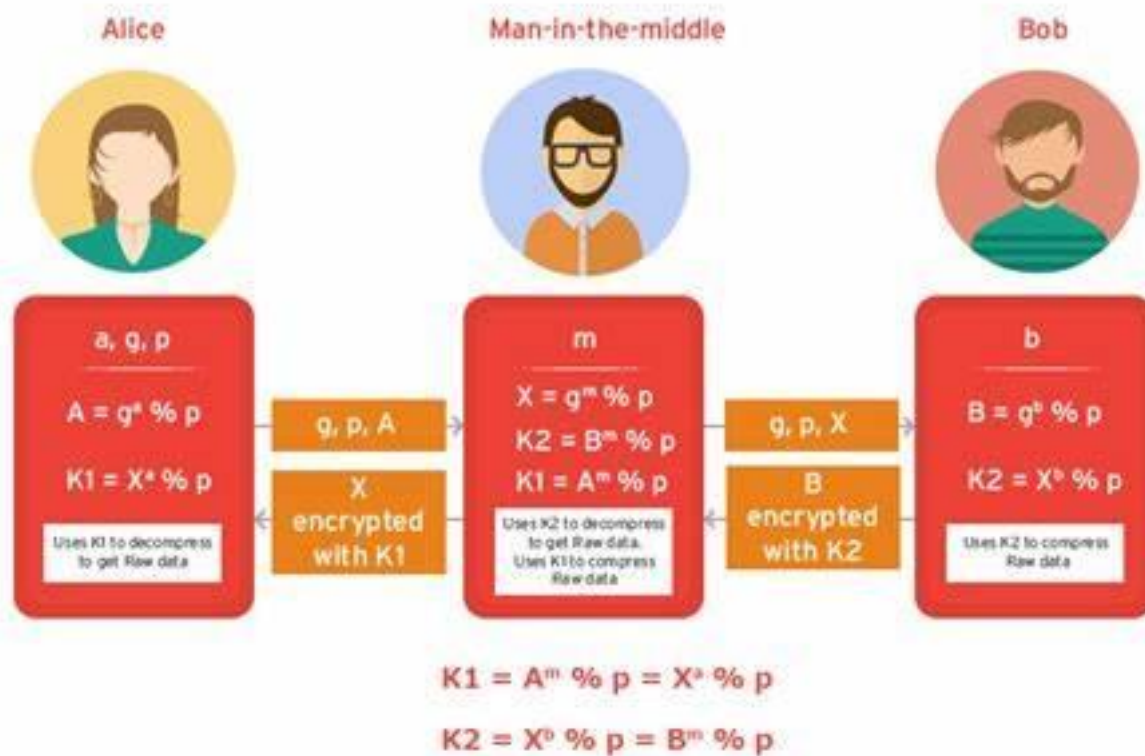


- Brute force try all possibility can still find the secret "a" but with a very heavy load.

<https://www.zhihu.com/question/35146790>

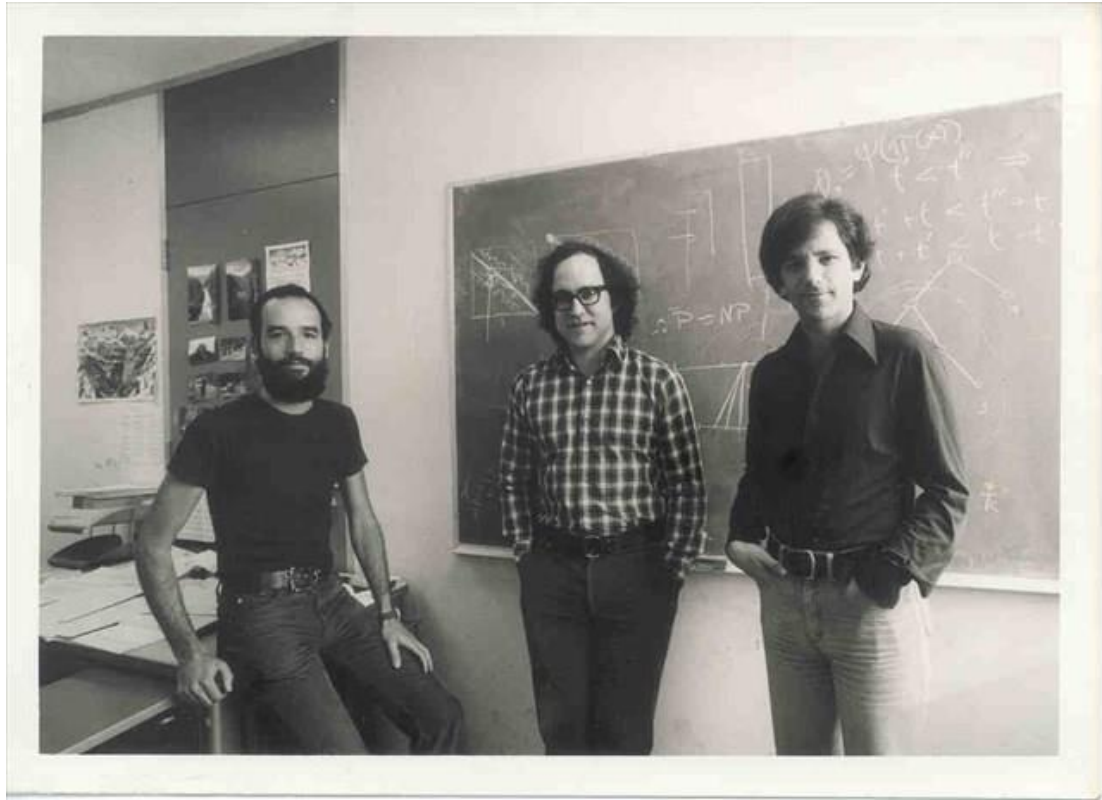
<https://math.stackexchange.com/questions/1580583/modular-exponentiation-commutativity-in-diffie-hellman>

Are we safe now?



- Only by D-H algorithm, **NO** way to make sure the public key is really from Alice or Bob

RSA asymmetric algorithm



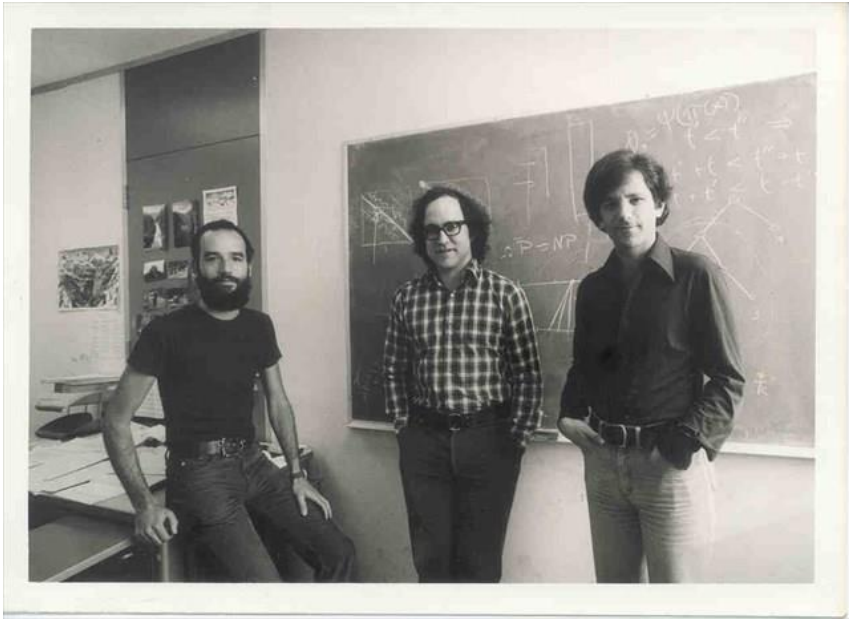
Rivest, Shamir and Adleman @1977

How RSA Encryption Works



- Typically, it uses key pairs. Keep one privately and public another that everyone can access it.
- Most information are public (assume everyone can access the ciphered text and public key)
- There is **theoretical difficulty** to deduce the secret from the public information

RSA algorithm



Rivest, Shamir and Adleman @1977

- **Still about the $\wedge$ and $\%$**
- **But in a different way to apply the math**

Condtions:

- $g < N$, to find the proper d , e and N meet **$g^{de} \bmod N = g$**
- Let put d , e and N as two keys: (d, N) and (e, N)
- So, we can make the following operations:
- $g^e \bmod N = \mathbf{C}$, and $C^d \bmod N = \mathbf{g^{de} \bmod N} = g$ (process C)
- $g^d \bmod N = \mathbf{S}$, and $S^e \bmod N = \mathbf{g^{de} \bmod N} = g$ (process S)
- If the results of intermediate steps: C and S are very difficult to be decrypt without the correct keys and it is difficult to deduce the (d, N) from (e, N) . Then the C and S can be public and no one can know the original g .

To find $\rightarrow g^{de-1} \bmod N = 1$

Euler's totient function & Fermat's little theorem

- **Euler's Totient function** $\varphi(x)$ for an input x is the count of numbers in
 - $\{1, 2, 3, \dots, n\}$ that are relatively prime to x .
 - When x is prime, then $\varphi(x) = x - 1$
 - When m and n are prime, then $\varphi(m*n) = (m - 1) * (n - 1)$
- **Euler theorem** $g^{\varphi(p)} \bmod p = 1$ (g could be any number)
- **So**, $g^{k\varphi(p)} \bmod p = 1$ (k is any integer number) **then** $g^{k\varphi(p)+1} \bmod p = g$,
- Now, we let $N = m*n$ (m and n are prime) and $d = k\varphi(m*n)+1$
- N is very easy to calculate and we normally choose a small number for e (i.e. 5, 17, 257, 65537) and deduce the d to have the key pairs: **(e, N)** and **(d, N)** as a key pair.

A sample of RSA keys



- let **$N = m * n$** (m and n are prime)
and **$de = k\phi(m * n) + 1$**

(1) select $m = 29, n = 83$

(2) $N = 29 * 83 = 2407, \phi(N) = (29 - 1) * (83 - 1) = 2296$

(3) Select $e = 5$

(4) Choose $k = 4, d = (4 * 2296 + 1) / 5 = 1837$

(5) Key pair is $\{5, 2407\}$ and $\{1837, 2407\}$

A test:

Original info: 199

Encryption it to $C = (199^5) \bmod 2407 = 1557$

Decryption C: $(1557^{1837}) \bmod 2407 = 199$

Crack RSA keys

- For key pairs $\{e, N\}$ and $\{d, N\}$, $\{e, p\}$ is public
- Since $N=m*n$ and $d*e = k\phi(m*n)+1$
- e and N are public
- If N can be divided into $m*n$ then d can be found easily
- The bigger the number, the difficult to factorize it by brute force. Nowadays 1024 bits are not safe due to the increasing calculation capability. At least it needs 2048 bits.
- Quantum computing would make it totally unsafe.

https://www.ruanyifeng.com/blog/2013/07/rsa_algorithm_part_two.html

12301866845301177551304949
58384962720772853569595334
79219732245215172640050726
36575187452021997864693899
56474942774063845925192557
32630345373154826850791702
61221429134616704292143116
02221240479274737794080665
351419597459856902143413

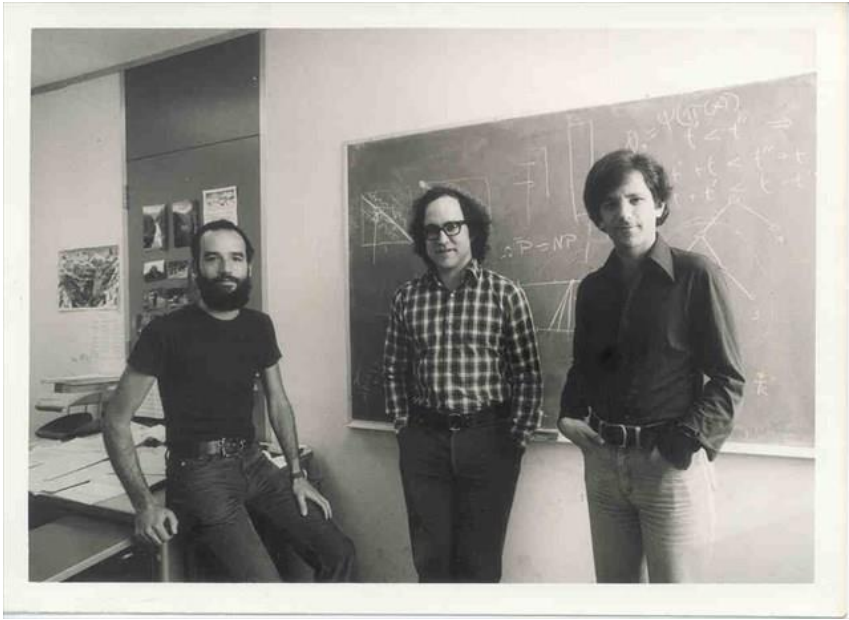
=

33478071698956898786044169
84821269081770479498371376
85689124313889828837938780
02287614711652531743087737
814467999489

×

36746043666799590428244633
79962795263227915816434308
76426760322838157396665112
79233373417143396810270092
798736308917

RSA applications



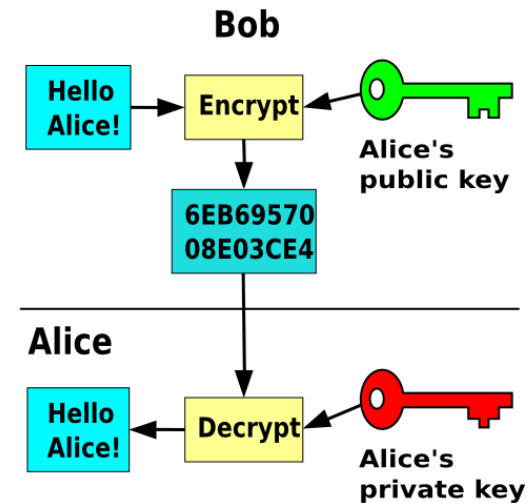
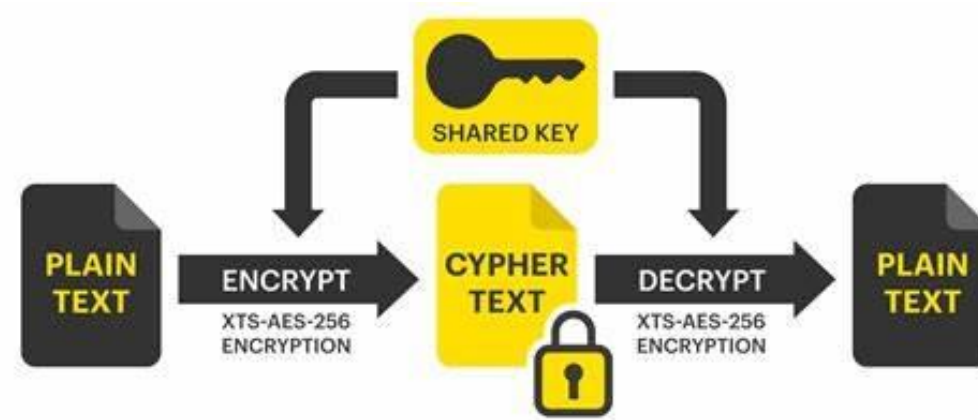
Rivest, Shamir and Adleman @1977

- Still about the $\wedge$ and $\%$
- But in a different way to apply the math

Two application approaches:

- For any application, it will keep (d, N) privately as private key and public the corresponding (e, N) as public key.
- Cipher (Process C): let others using the public key to encrypt, only the person has private key can decrypt.
 - Encryption: $g^e \bmod N = \text{C}$,
 - Decryption: $C^d \bmod N = g^{de} \bmod N = g$
- Digital signature (Process S): only the one has private key can deduce S, others can verify the S is deduced from g by the public key.
 - Sing: $g^d \bmod N = \text{S}$,
 - Verify: $S^e \bmod N = g^{de} \bmod N = g$

Symmetric and Asymmetric encryption



- For asymmetric one, everyone can know the public key, no afraid of middle man.

Verify original by signature

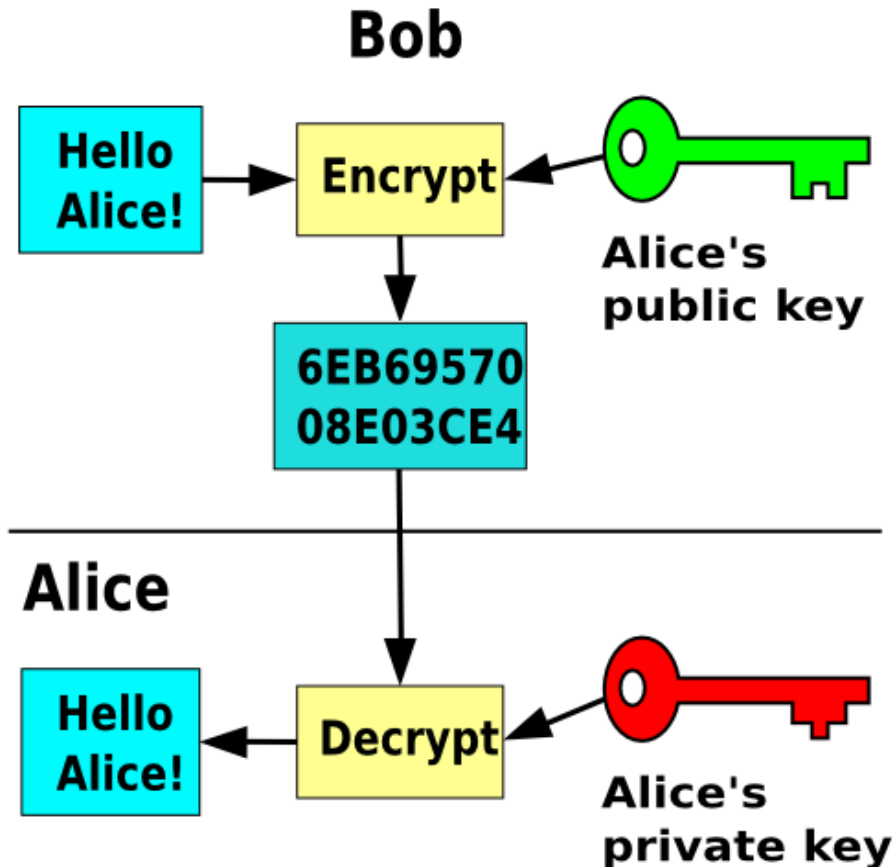


Digital signature



- Signature as a picture is very fragile

RSA digital signature



- It can be done in a reverse way
- Alice use the private key to encrypt the original message.
- The receiver can decrypt it with Alice's public key to verify its original.
- Problems:
 - The text cannot be read in plain text
 - RSA calculation is heavy for CPU.

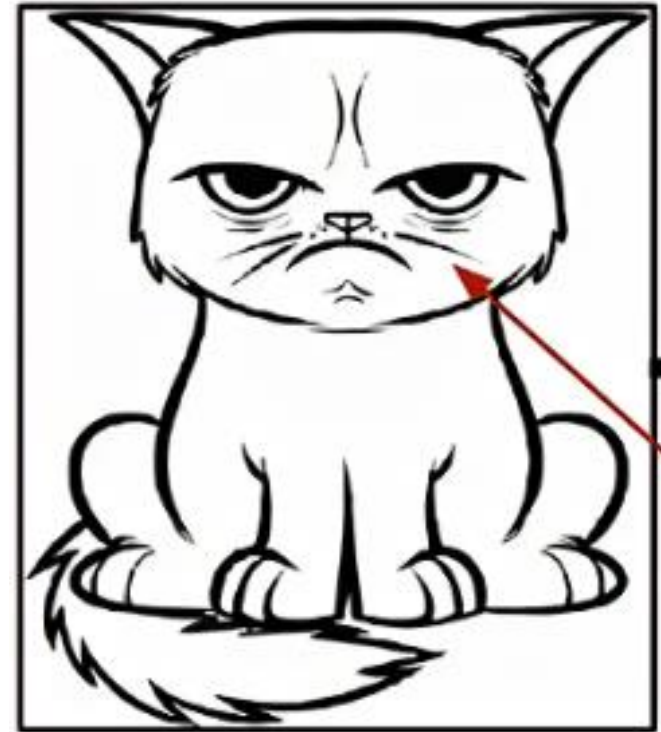
Hash



A dish of cooked meat cut into small pieces and cooked again.

<https://www.merriam-webster.com/dictionary/hash>

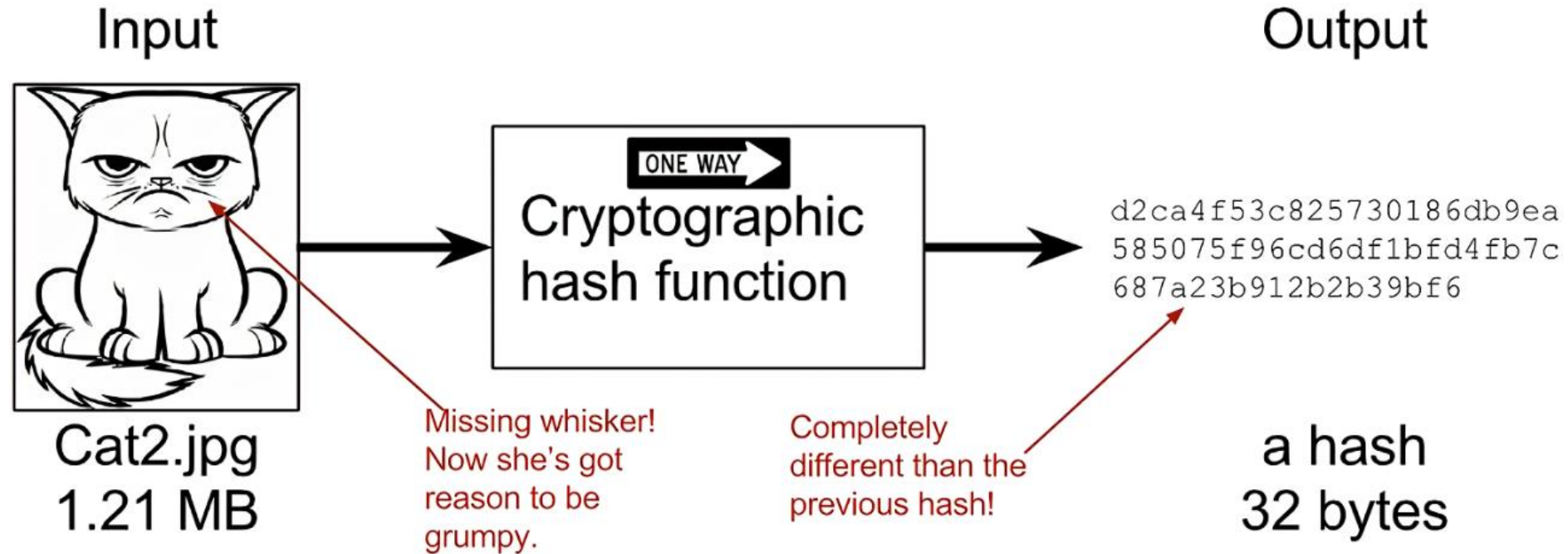
Hash Encoding



Cat2.jpg

Hash is the message digest
Mathematical function that converts an input of arbitrary length into an output of a fixed length.

Hash: message digest

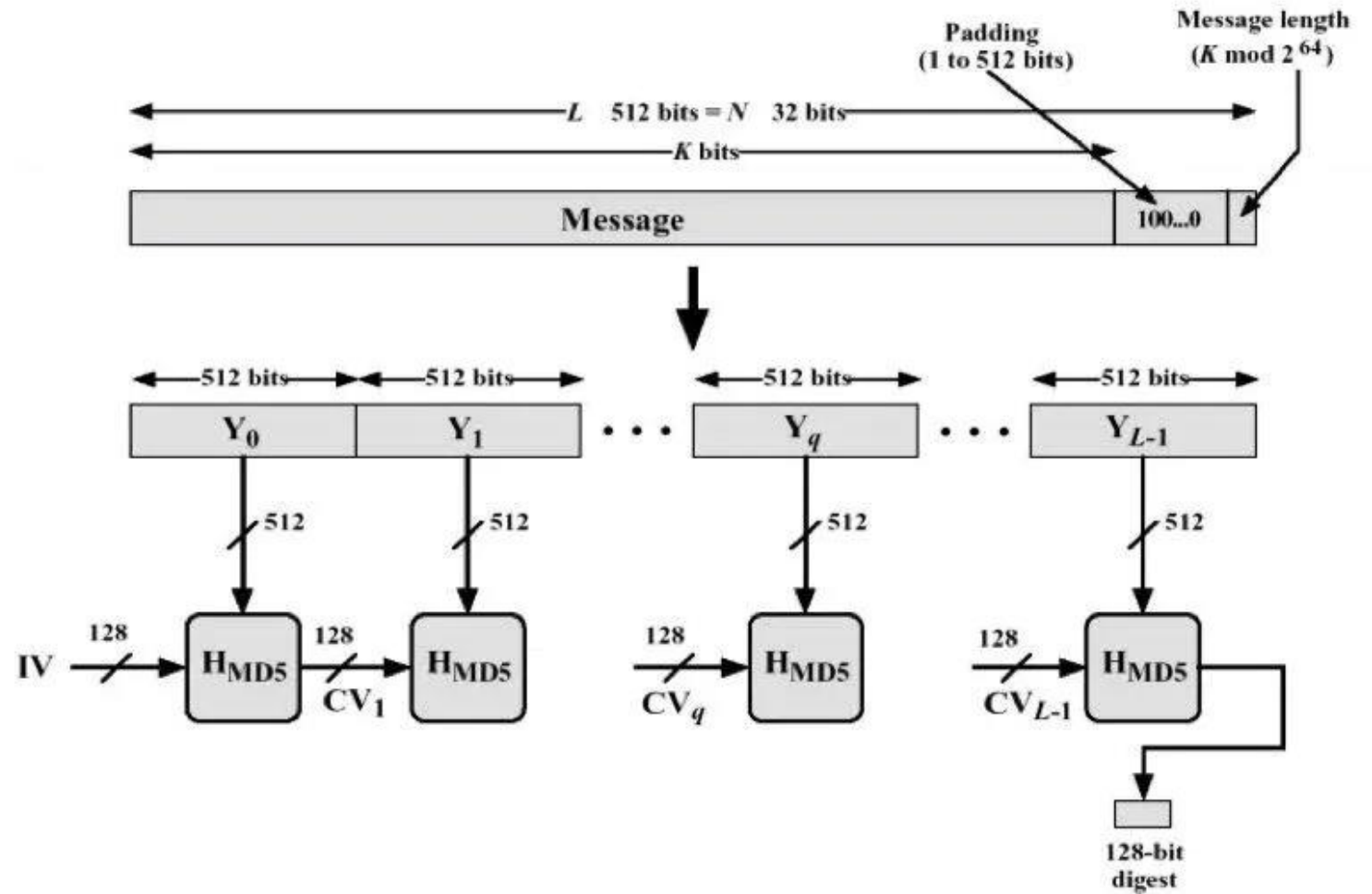


- By design, Hash is **ONLY** one way, "**NO**" way to **decode** it back
- It should supply enough space for different data
- A small change in data should have total different hash value

Hash: MD5

h_0	01	23	45	67
h_1	89	AB	CD	EF
h_2	FE	DC	BA	98
h_3	76	54	32	10

```
1 // Initialize variables - simple count in nibbles:
2 h0 = 0x67452301;
3 h1 = 0xefcdab89;
4 h2 = 0x98badcfe;
5 h3 = 0x10325476;
```



- A widely used hash algorithm

<https://www.jianshu.com/p/2d2debf78f69>

SHA series

$H_0^{(0)} = 67452301$
 $H_1^{(0)} = \text{efcdab89}$
 $H_2^{(0)} = 98badcfe$
 $H_3^{(0)} = 10325476$
 $H_4^{(0)} = \text{c3d2e1f0.}$

3. For $t = 0$ to 79:

$\{$
 $T = \text{ROTL}^5(a) + f_t(b, c, d) + e + K_t + W_t$
 $e = d$
 $d = c$
 $c = \text{ROTL}^{30}(b)$
 $b = a$
 $a = T$
 $\}$

$$K_t = \begin{cases} 5a827999 & 0 \leq t \leq 19 \\ 6ed9eba1 & 20 \leq t \leq 39 \\ 8f1bbcdc & 40 \leq t \leq 59 \\ ca62c1d6 & 60 \leq t \leq 79. \end{cases} \quad (4.14)$$

Hash Viewer

CRC32	D96472E5	☒
SHA1	7B0041E50D621B7EFB45FFF8D09	☒
SHA256	FC3DA10B42BB81B3796AC6FFA6	☒
SHA384	08A3CCF189E9E55F2D811C54059	☒
SHA512	CDD23B8623A93346DA1A601BBF	☒
MD5	9FCA53A5FF8903277AAD8994750	☒

File Name: 0.txt
File Size: 9.05 KB

Verify ☐

☒ Uppercase
 ☒ Context menu
 ☒ TopMost

Select file or drag and drop

Hash: widely used for verifying the integrity

Windows, 64-bit

Windows, 64-bit

Current Version: 0.14.0.0 Boron Butterfly

SHA256 Hash (GUI):

c23c849a2a970370f2bf7e22aae34f633f51d8490b6130337e5fe9369
aff77e0

Windows, 64-bit Installer

Current Version: 0.14.0.0 Boron Butterfly

SHA256 Hash (GUI Installer):

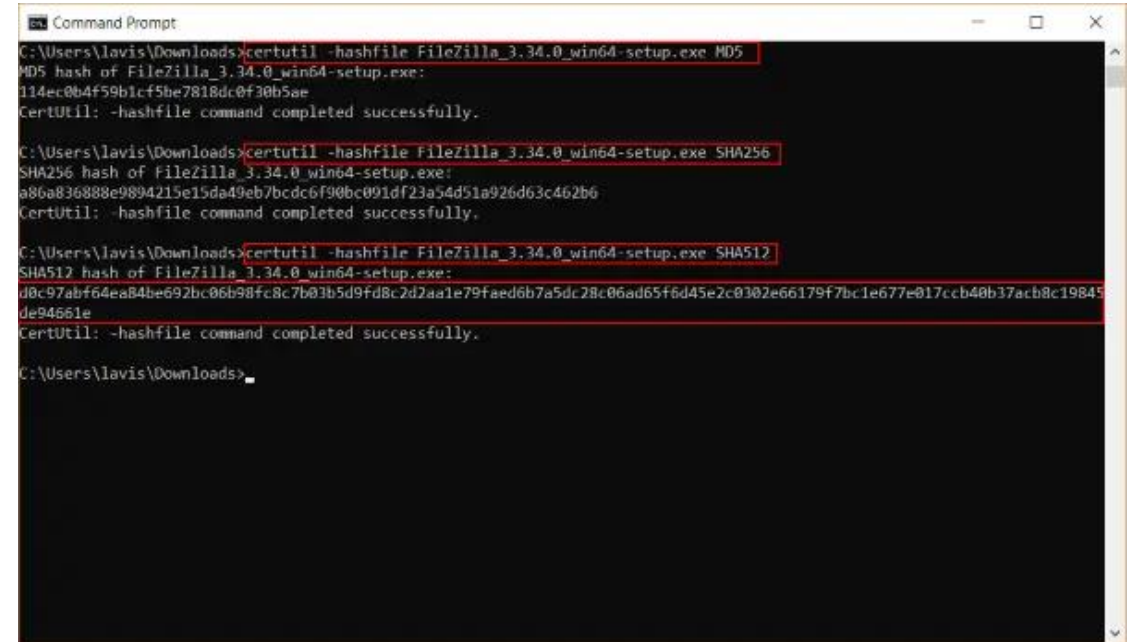
4c0ceb81361a4b69bfb6cd5a47ef94cfd2992796fb25ca3923aed51d6
fcd48bd

Windows, 64-bit Command-Line Tools Only

Current Version: 0.14.0.2 Boron Butterfly

SHA256 Hash (CLI):

ee68ef8d4b3309cca9e68ee272919c222021804e493c2549c910ef370
dfe595a



```
Command Prompt
C:\Users\lavis\Downloads>certutil -hashfile FileZilla_3.34.0_win64-setup.exe MD5
MD5 hash of FileZilla_3.34.0_win64-setup.exe:
114ec0b4f59b1cf5be7818dc0f30b5ae
CertUtil: -hashfile command completed successfully.

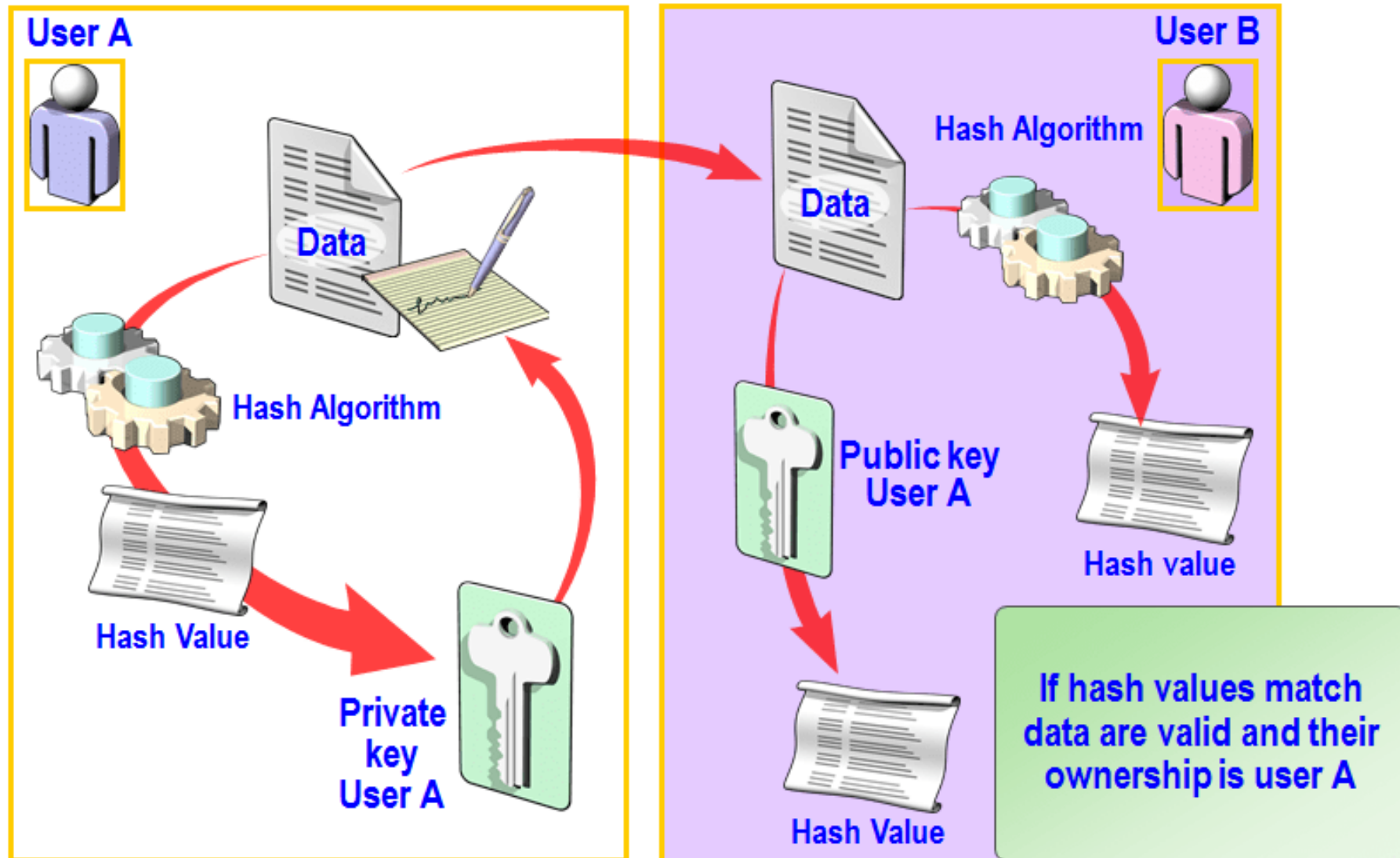
C:\Users\lavis\Downloads>certutil -hashfile FileZilla_3.34.0_win64-setup.exe SHA256
SHA256 hash of FileZilla_3.34.0_win64-setup.exe:
a86a836888e9894215e15da49eb7bcd6f90bc091df23a54d51a926d63c462b6
CertUtil: -hashfile command completed successfully.

C:\Users\lavis\Downloads>certutil -hashfile FileZilla_3.34.0_win64-setup.exe SHA512
SHA512 hash of FileZilla_3.34.0_win64-setup.exe:
d0c97abf64ea84be692bc06b98fc8c7b03b5d9fd8c2d2aale79faed6b7a5dc28c06ad65f6d45e2c0302e66179f7bc1e677e017ccb40b37acb8c19845
de94661e
CertUtil: -hashfile command completed successfully.

C:\Users\lavis\Downloads>
```

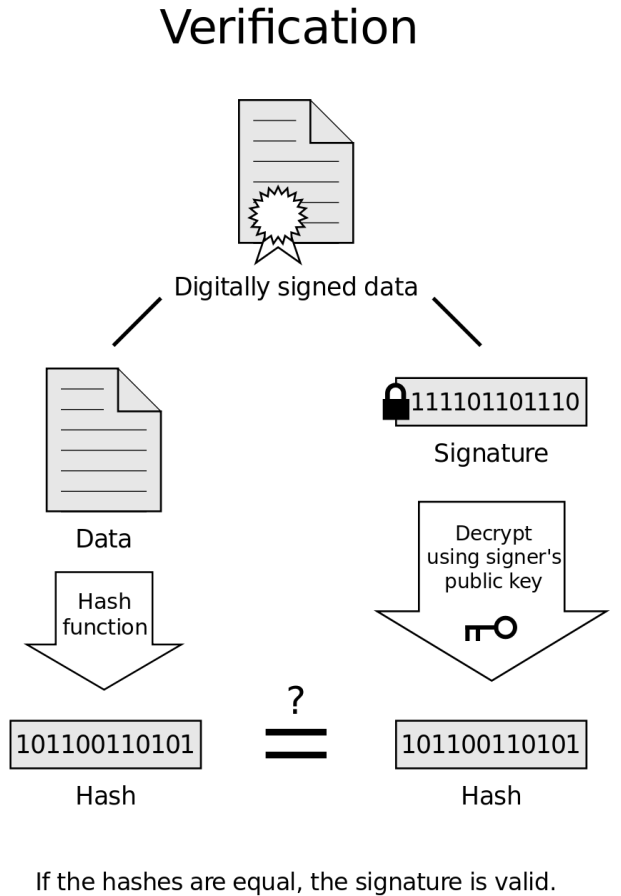
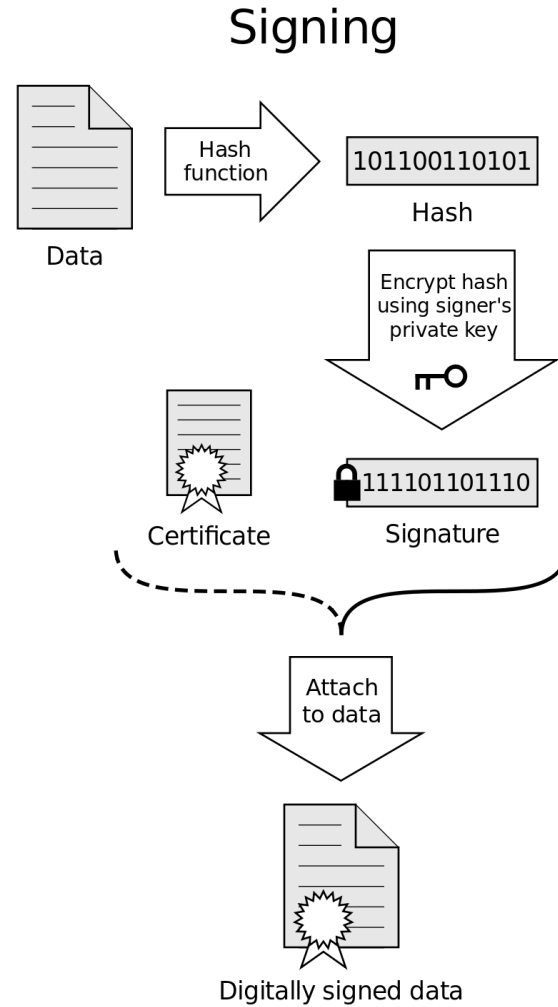
- Use the same hash function to calculate the hash value and compare with the published one.

Digital signature



- Only need to sign the HASH but not the full text

Digital signature vs. traditional stamp



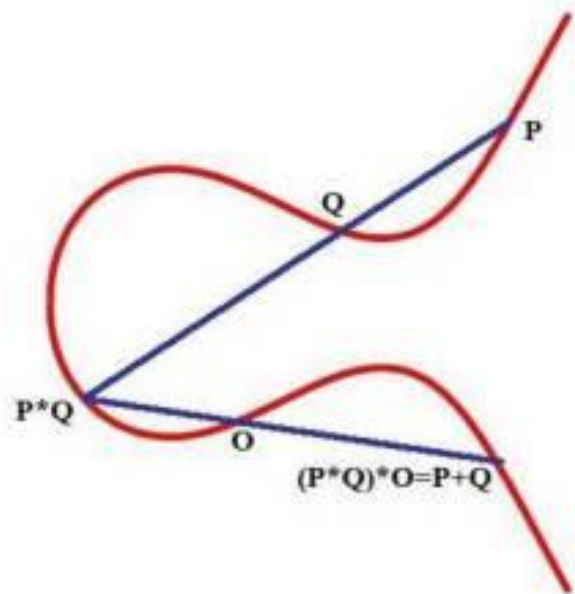
- RSA can be used for digital signature application.

More asymmetric **keys** algorithms



- Basic math challenge is similar to DH.
- There are very easy “add, minus and plus” but damned hard “**divide**”.

ECC: **Elliptic** Curve Cryptography



$$y^2 = x^3 + ax + b$$

secp256k1标准曲线的领域参数如下:

```
1 | p = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEFFFFFC2F
2 | a = 0
3 | b = 7
4 | xG = 0x79BE667EF9DCBBAC55A06295CE870B07029BFCDB2DCE28D959F2815B16F81798
5 | yG = 0x483ADA7726A3C4655DA4FBFC0E1108A8FD17B448A68554199C47D08FFB10D4B8
6 | n = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141
7 | h = 1
```

- Much more complex math theory.
- The parameters need to be chosen for a specific curve

<https://www.zhihu.com/question/29549792>

Chinese algorithms of asymmetric encryption



国家密码管理局

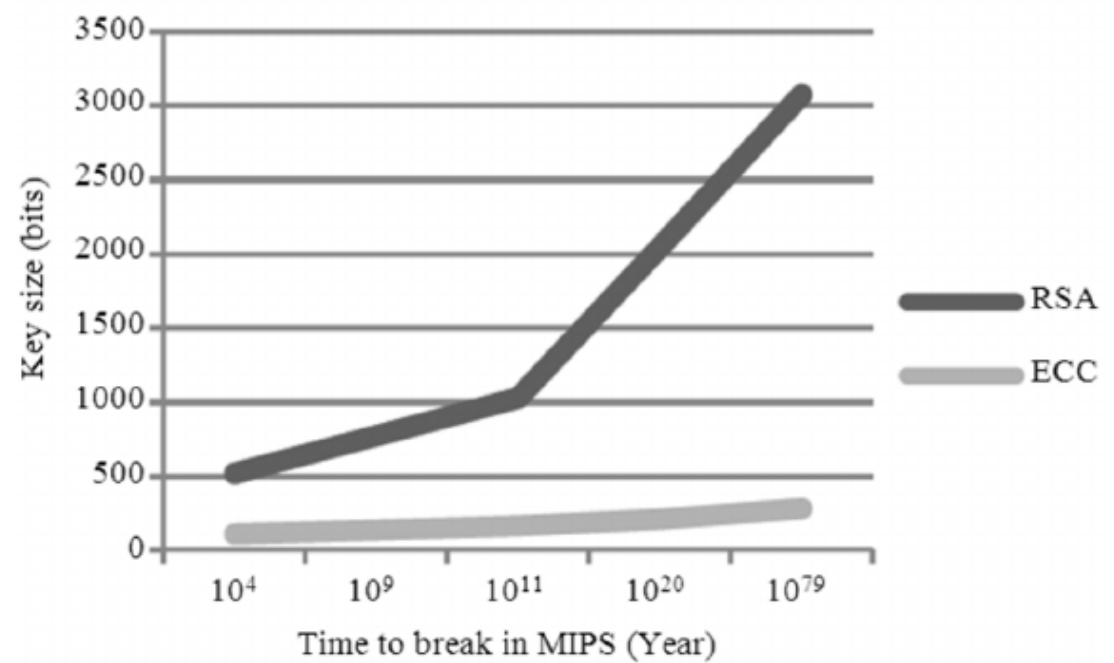
WWW.SCA.GOV.CN

SM2标准椭圆曲线叫做sm2p256v1，具体领域参数如下：

```
1 | p = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF00000000FFFFFFFFFFFFFFFF
2 | a = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF00000000FFFFFFFFFFFFFFFC
3 | b = 0x28E9FA9E9D9F5E344D5A9E4BCF6509A7F39789F515AB8F92DDBCBD414D940E93
4 | xG = 0x32C4AE2C1F1981195F9904466A39C9948FE30BBFF2660BE1715A4589334C74C7
5 | yG = 0xBC3736A2F4F6779C59BDCEE36B692153D0A9877CC62A474002DF32E52139F0A0
6 | n = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF7203DF6B21C6052B53BBF40939D54123
7 | h = 1
```

- SM2 is an ECC with a group of specific parameters

ECC: **Elliptic** Curve Cryptography



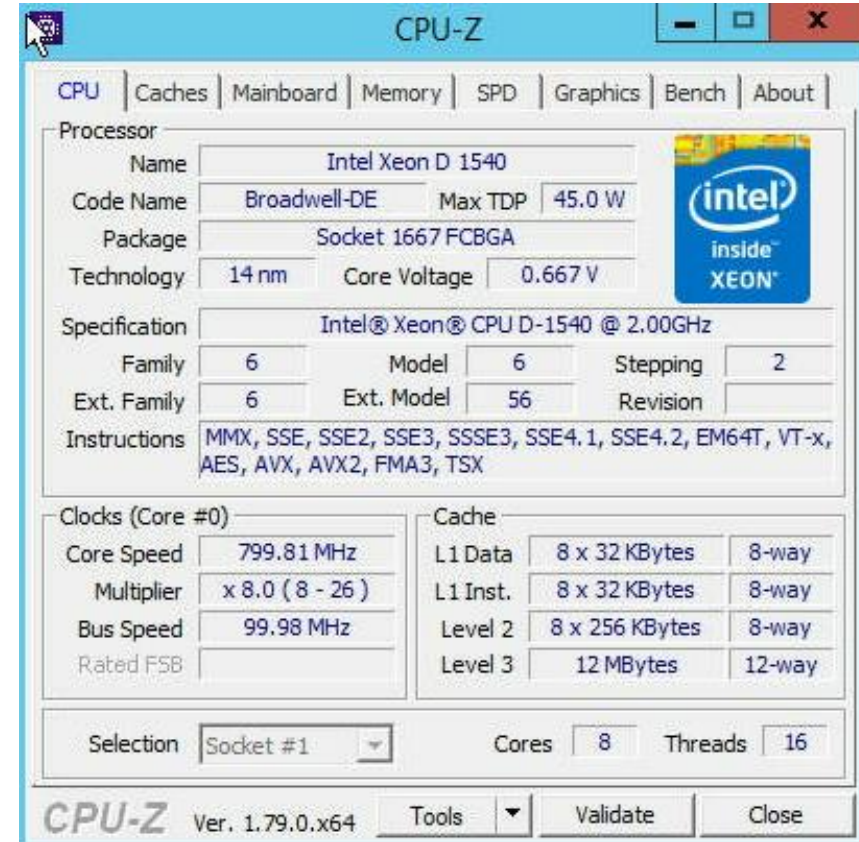
	Signature generation (ms)	Signature verification (ms)
ECDSA- $F_{p=160}$	41.14	39.81
RSA-1024	81.22	5.05
ECDSA- $F_{p=192}$	45.59	42.92
RSA-1536	504.06	15.19
ECDSA- $F_{p=224}$	48.76	43.12
RSA-2048	1254.42	78.60

- Relative short key to achieve the same security against brute force cracking.
- So, ECC is faster than RSA for the same security level.
- Up-to-date application prefer ECC.
- Blockchains like **Bitcoin** using ECC encryption.

<https://www.zhihu.com/question/29549792>

Cipher cost for different algorithms

	Encryption Technique	Using Matlab	Using FPGA
Encryption time in seconds	DES	400	90
	RSA	300	75
	Blowfish	100	40
	128- bits AES	5	3
	192- bits AES	7	5
	256- bits AES	9	8
Decryption time in seconds	DES	390	250
	RSA	350	180
	Blowfish	120	85
	128- bits AES	8	5
	192- bits AES	9.5	7
	256- bits AES	13	10



- Since RSA need a relatively long key, the calculation is more time consuming
- Modern CPUs have build-in AES algorithm (hardware optimized)

Short summary of Symmetric and Asymmetric

Table 1. Security Comparison for Various Algorithm-key Size Combinations (Source: NSA) ⁽⁷⁾

Security Bits	Symmetric Encryption Algorithm	Minimum Size (bits) of Public Keys	
		RSA	ECC
80	Skipjack	1024	160
112	3DES	2048	224
128	AES-128	3072	256
192	AES-192	7680	384
256	AES-256	15360	512

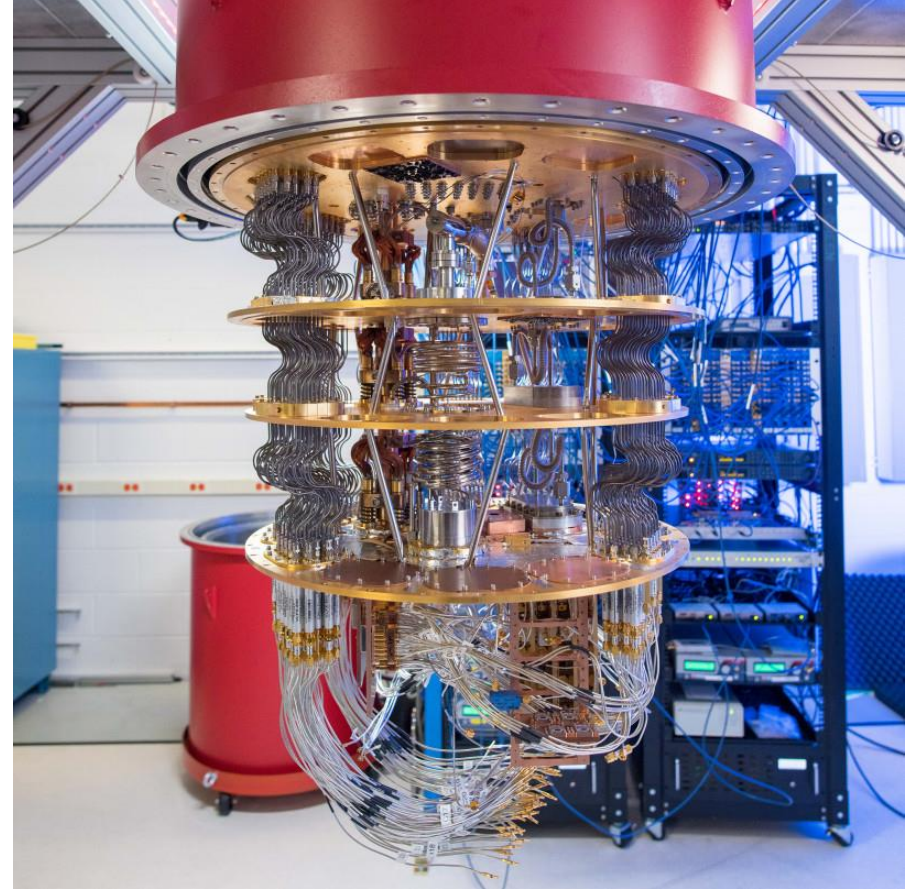
- The security of different algorithms **CANNOT** be compared directly.
- The comparison is based on the similar MIPS years for brute force cracking.
- Maybe surprise, symmetric encryption **is** more **difficult** to break than asymmetric one for same key length.
- The disadvantage of symmetric encryption is how to **SHARE** the keys between Alice and Bob safely.
- All keys **CAN** be broken. The main issue is about "computing resource", does it worth to be broken?
- New technology may make certain algorithm no longer secure.
- One good application always **combines** the advantages of different algorithms.

<https://www.netburner.com/learn/comparing-rsa-and-ecc-encryption/>

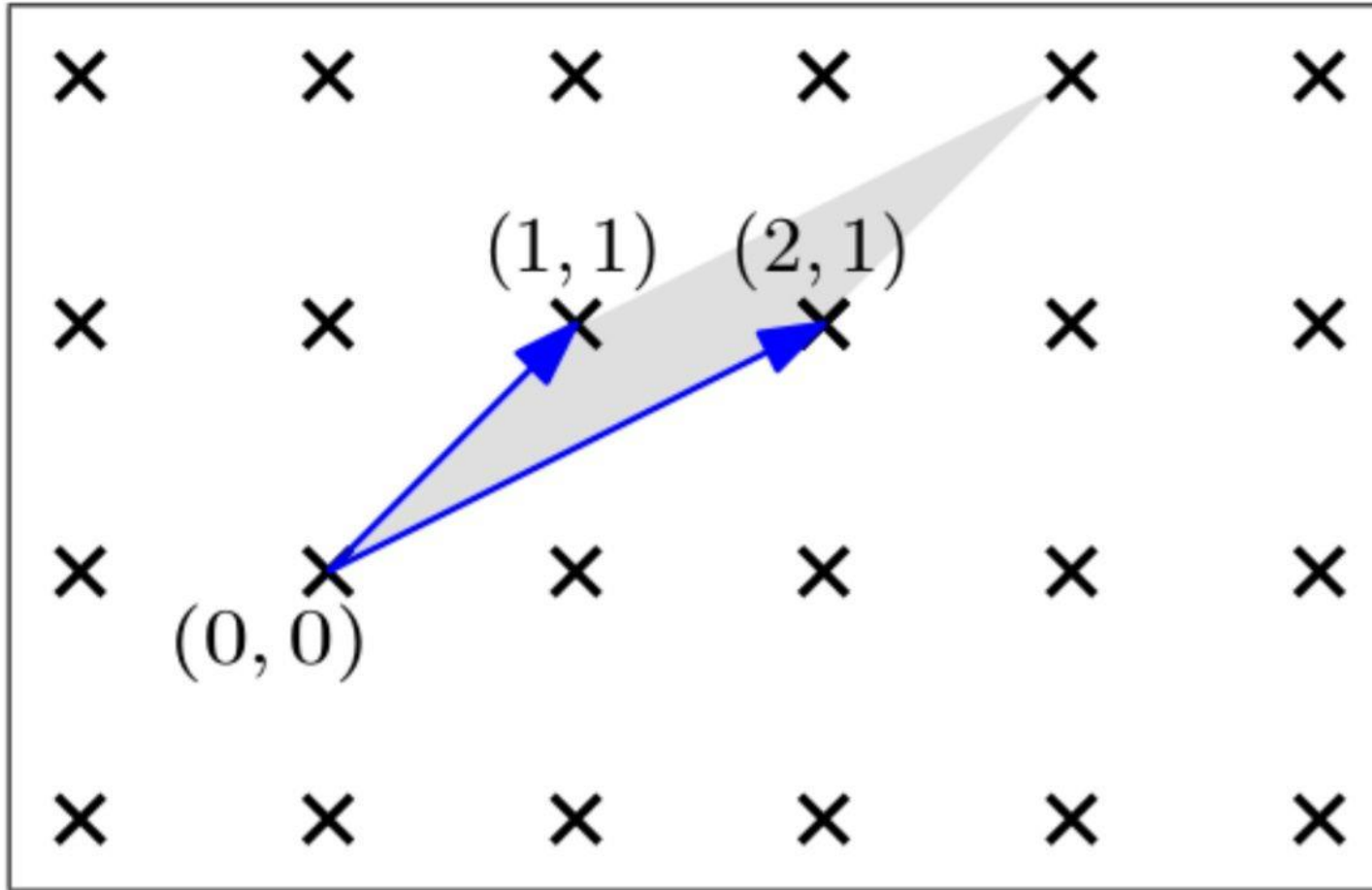
Shor algorithm by Quantum computer

$$F(|\phi\rangle) = \frac{1}{\sqrt{4}} \left[\begin{aligned} &\frac{1}{\sqrt{8}} \left(|0\rangle + e^{i\frac{2\pi}{8}} |1\rangle + e^{i\frac{2\pi}{8}2} |2\rangle + e^{i\frac{2\pi}{8}7} |7\rangle \right) \\ &+ \frac{1}{\sqrt{8}} \left(|0\rangle + e^{i\frac{2\pi 3}{8}} |1\rangle + e^{i\frac{2\pi 3}{8}2} |2\rangle + e^{i\frac{2\pi 3}{8}7} |7\rangle \right) \\ &+ \frac{1}{\sqrt{8}} \left(|0\rangle + e^{i\frac{2\pi 5}{8}} |1\rangle + e^{i\frac{2\pi 5}{8}2} |2\rangle + e^{i\frac{2\pi 5}{8}7} |7\rangle \right) \\ &+ \frac{1}{\sqrt{8}} \left(|0\rangle + e^{i\frac{2\pi 7}{8}} |1\rangle + e^{i\frac{2\pi 7}{8}2} |2\rangle + e^{i\frac{2\pi 7}{8}7} |7\rangle \right) \end{aligned} \right]$$

- Calculate the m and n from p
- So far, the public reported maximum number is 21.

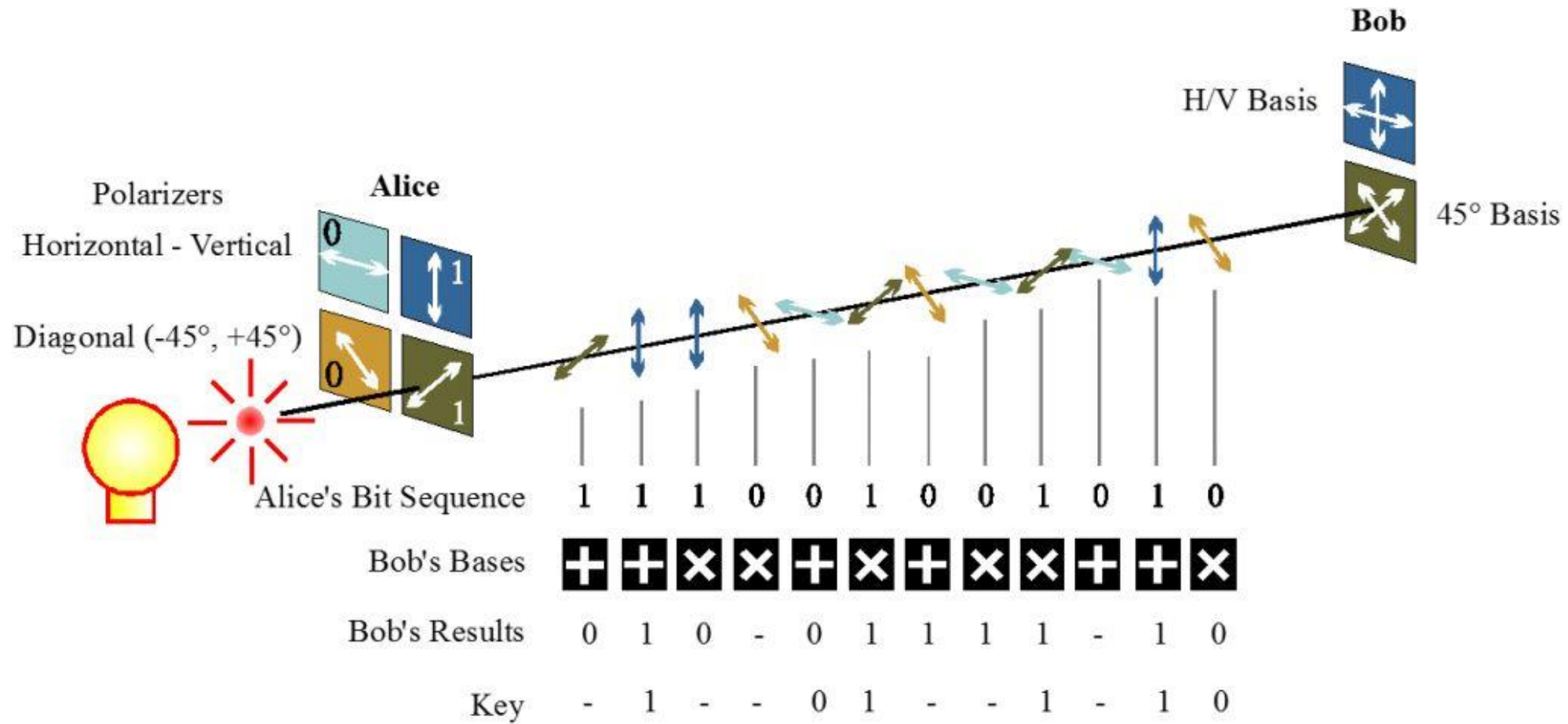


Post quantum algorithm: lattice encryption



(b) Another basis of $\mathbb{Z}^2$ 知乎 @孤雁伴月

Quantum key **may** against quantum computing!



<https://www.qtumist.com/post/2178>